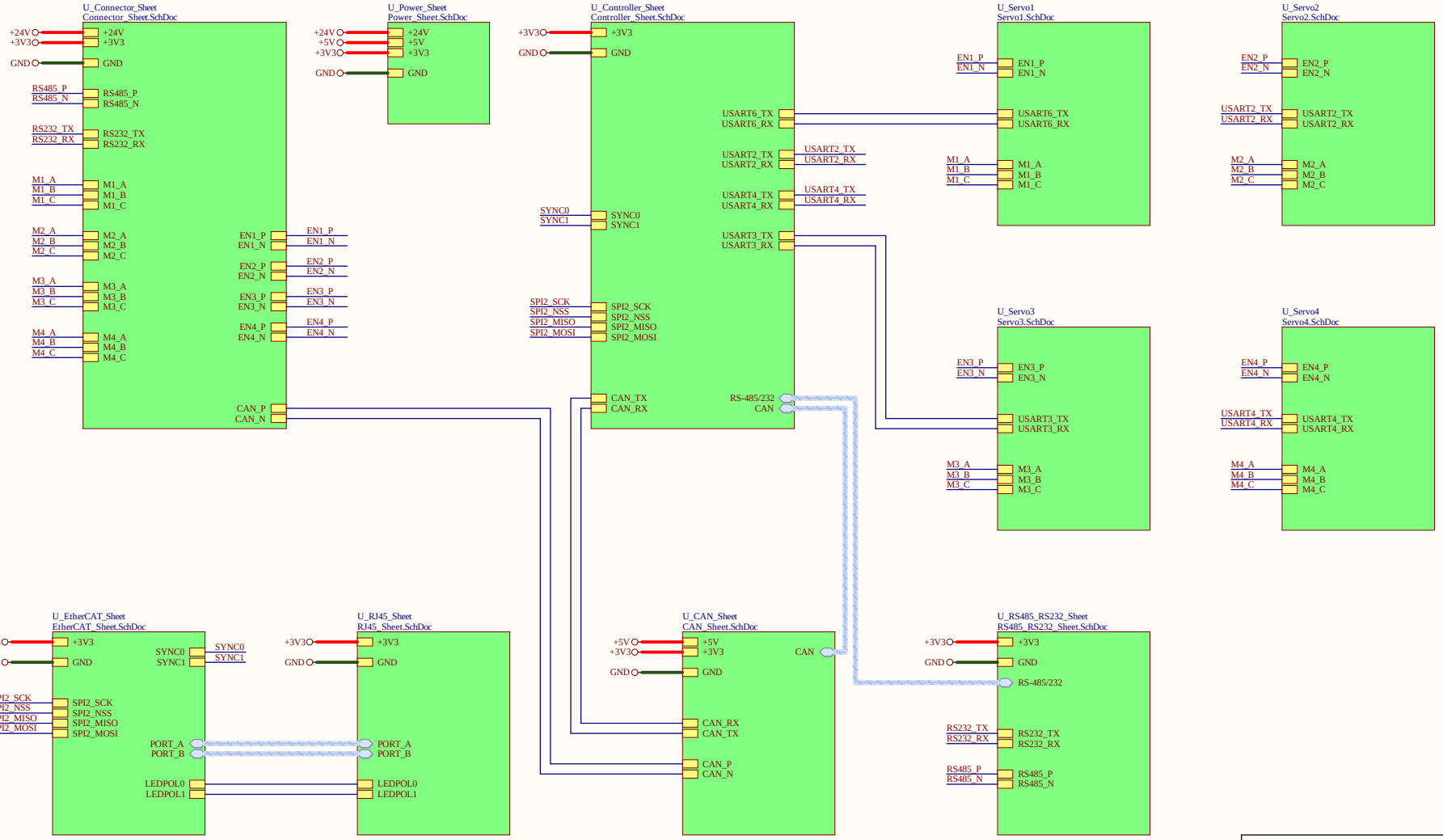
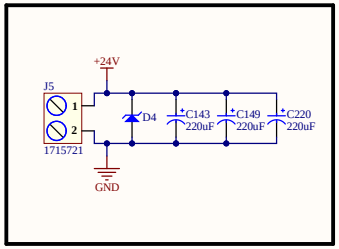


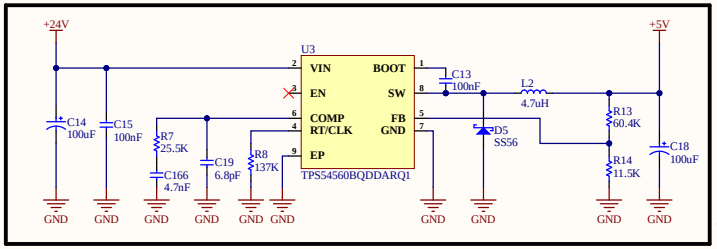
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Date: 8/9/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 32

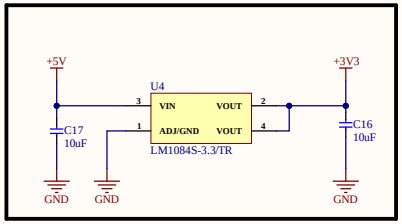
24V INPUT



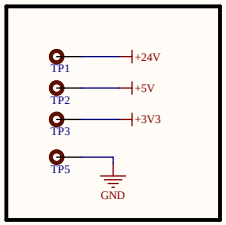
5V 5A BUCK



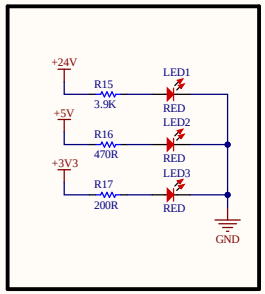
3.3V 3A LDO

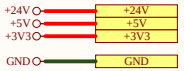


TEST POINTS



LEDs

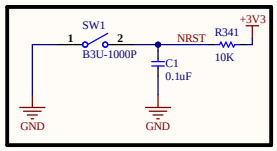




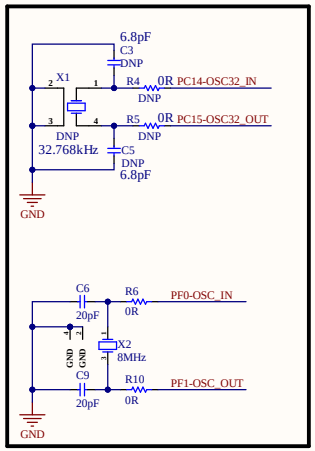
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Date: 8/8/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 32

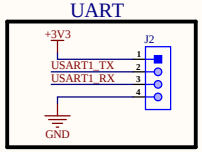
RESET



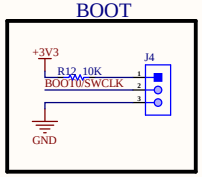
CRYSTAL



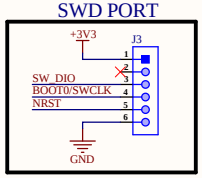
UART

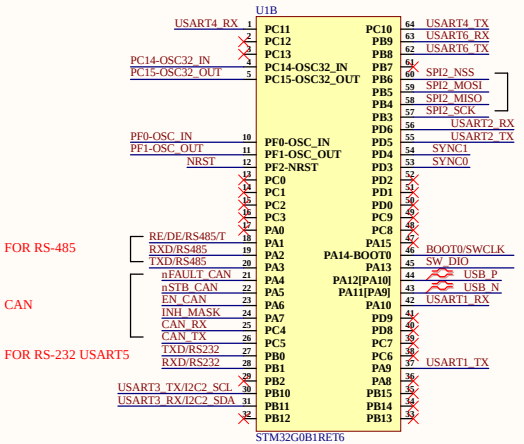


BOOT



SWD PORT



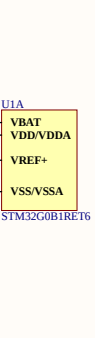


FOR RS-485

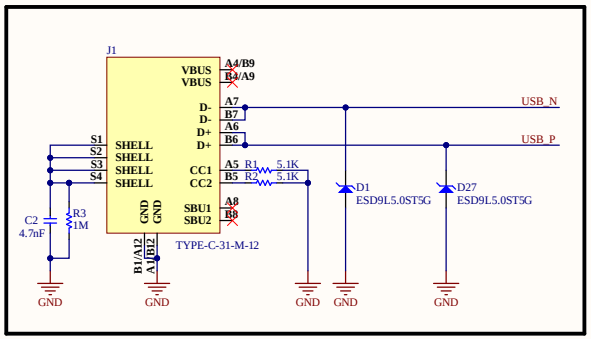
CAN

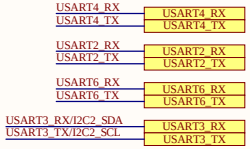
FOR RS-232 USART5

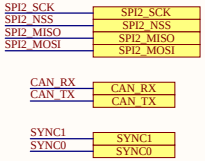
EtherCAT SPI

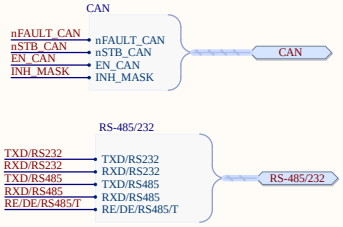


USB







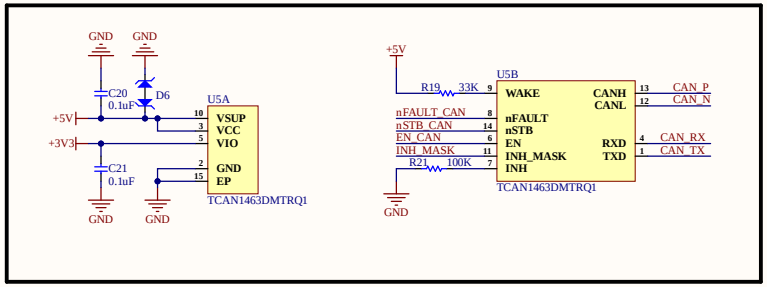




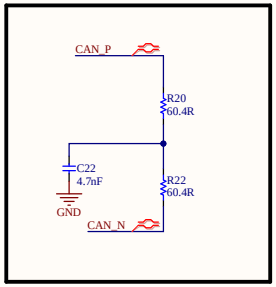
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Date: 8/8/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 32

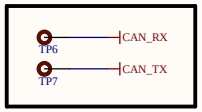
CAN BUS

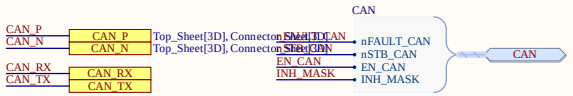


CAN TERMINATION



TESTPOINTS



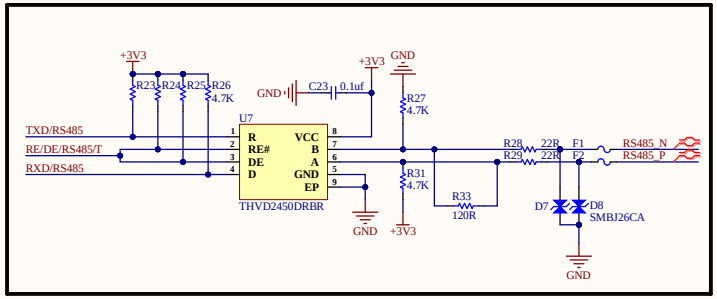




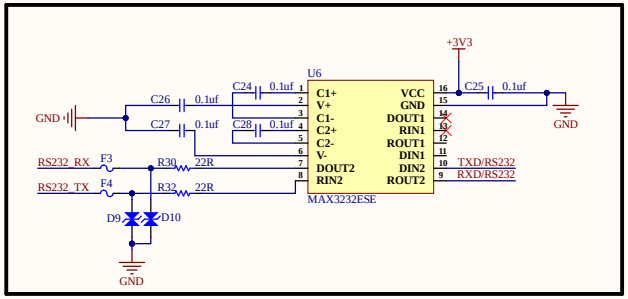
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Title: CAN_Sheet.SchDoc		
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Date: 8/9/2024	Page	Num: 4
Drawn By: Huzaifa Ali	Page	Total: 32

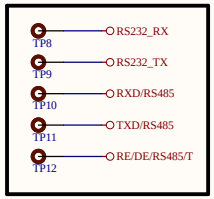
RS485 TO UART

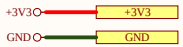


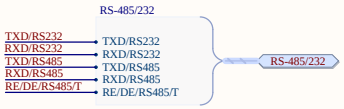
RS232 TO UART

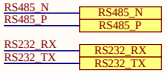


TESTPOINTS





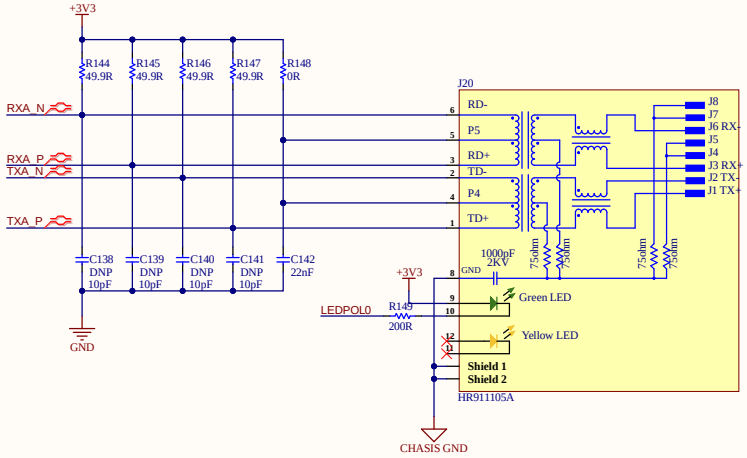




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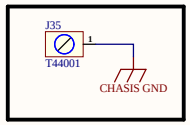
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Date: 8/9/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 32

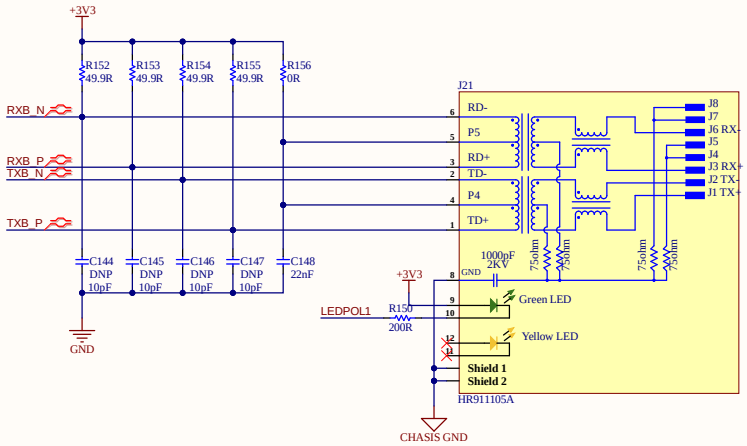
RJ45



INPUT

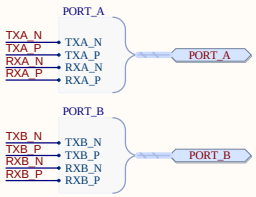
EARTH TERMINAL

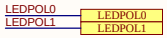




OUTPUT

LED1 (Green) = LINK/ACT
LED2 (Yellow) = SPEED

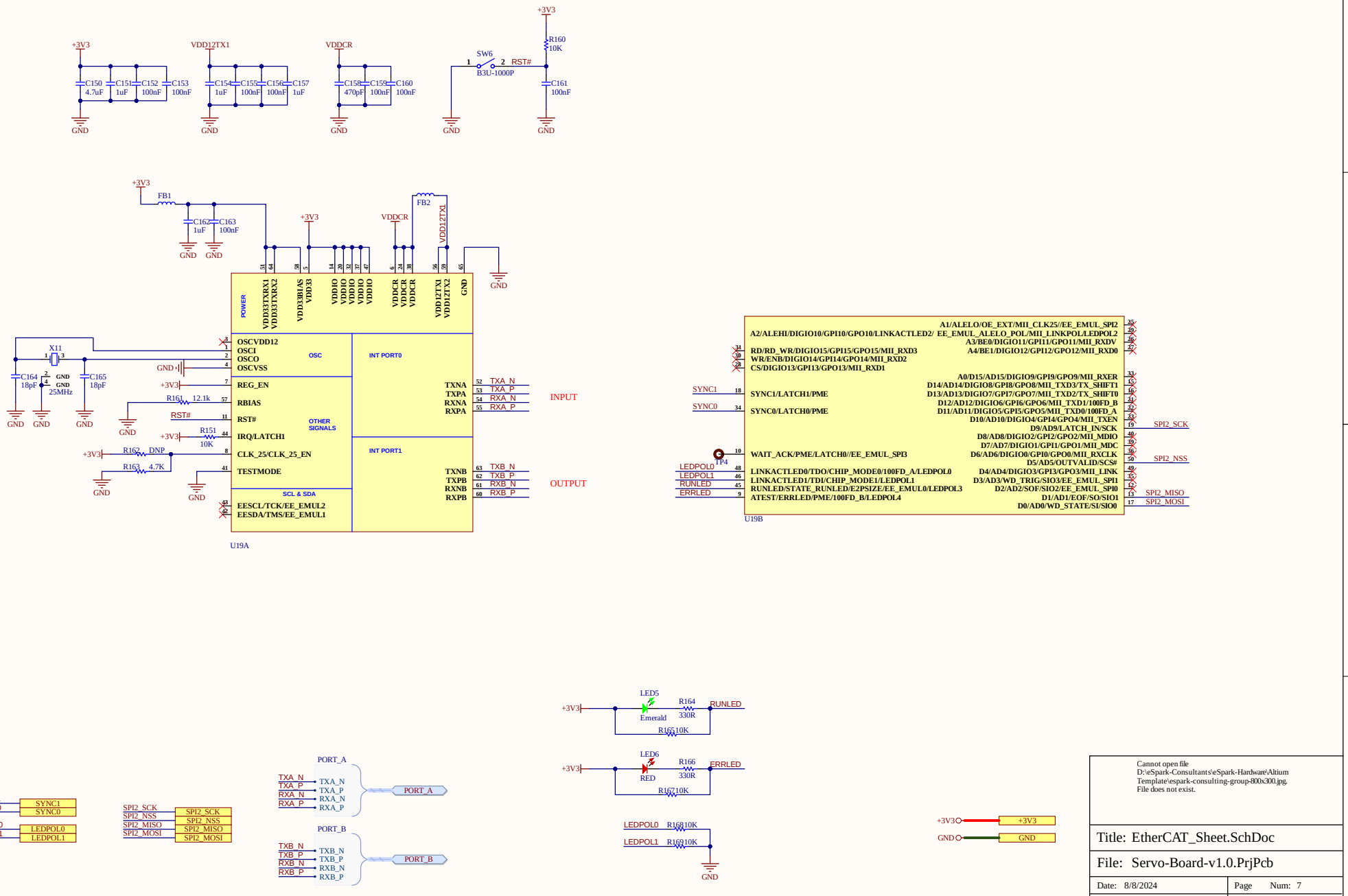




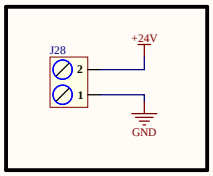


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Title: RJ45_Sheet.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/9/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 32


+3V3
VDD12TX1
VDDCR
C150 4.7uF
C151 1uF
C152 100nF
C153 100nF
C154 1uF
C155 100nF
C156 100nF
C157 1uF
C158 470pF
C159 100nF
C160 100nF
SW6
B3U-1000P
RST#
R160 10K
C161 100nF
GND
FB1
FB2
C162 1uF
C163 100nF
POWER
VDD33TXRX1
VDD33TXRX2
VDD33BIAS
VDD33
VDDIO
VDDCR
VDD12TX1
VDD12TX2
OSCVDD12
OSCI
OSCO
OSCVSS
OSC
REG_EN
RBIAS
RST#
IRQ/LATCH1
CLK_25/CLK_25_EN
TESTMODE
OTHER SIGNALS
SCL & SDA
EESCL/TCK/EE_EMUL2
EESDA/TMS/EE_EMUL1
INT PORT0
INT PORT1
TXNA
TXPA
RXNA
RXPA
TXNB
TXPB
RXNB
RXPB
X11
C164 18pF
C165 18pF
25MHz
R161 12.1k
R151 10K
R162 DNP
R163 4.7K
INPUT
OUTPUT
U19A
A1/ALELO/OE_EXT/MII_CLK25//EE_EMUL_SPI2
A2/ALEHI/DIGIO10/GPI10/GPO10/LINKACTLED2/
EE_EMUL_ALELO_POL/MII_LINKPOL/LEDPOL2
A3/BE0/DIGIO11/GPI11/GPO11/MII_RXDV
RD/RD_WR/DIGIO15/GPI15/GPO15/MII_RXD3
A4/BE1/DIGIO12/GPI12/GPO12/MII_RXD0
WR/ENB/DIGIO14/GPI14/GPO14/MII_RXD2
CS/DIGIO13/GPI13/GPO13/MII_RXD1
A0/D15/AD15/DIGIO9/GPI9/GPO9/MII_RXER
D14/AD14/DIGIO8/GPI8/GPO8/MII_TXD3/TX_SHIFT1
SYNC1/LATCH1/PME
D13/AD13/DIGIO7/GPI7/GPO7/MII_TXD2/TX_SHIFT0
D12/AD12/DIGIO6/GPI6/GPO6/MII_TXD1/100FD_B
SYNC0/LATCH0/PME
D11/AD11/DIGIO5/GPI5/GPO5/MII_TXD0/100FD_A
D10/AD10/DIGIO4/GPI4/GPO4/MII_TXEN
D9/AD9/LATCH_IN/SCK
D8/AD8/DIGIO2/GPI2/GPO2/MII_MDIO
D7/AD7/DIGIO1/GPI1/GPO1/MII_MDC
WAIT_ACK/PME/LATCH0//EE_EMUL_SPI3
D6/AD6/DIGIO0/GPI0/GPO0/MII_RXCLK
D5/AD5/OUTVALID/SCS#
LINKACTLED0/TDO/CHIP_MODE0/100FD_A/LEDPOL0
D4/AD4/DIGIO3/GPI3/GPO3/MII_LINK
LINKACTLED1/TDI/CHIP_MODE1/LEDPOL1
D3/AD3/WD_TRIG/SIO3/EE_EMUL_SPI1
RUNLED/STATE_RUNLED/E2PSIZE/EE_EMUL0/LEDPOL3
D2/AD2/SOF/SIO2/EE_EMUL_SPI0
ATEST/ERRLED/PME/100FD_B/LEDPOL4
D1/AD1/EOF/SO/SIO1
D0/AD0/WD_STATE/SI/SIO0
TP4
U19B
SPI2_SCK
SPI2_NSS
SPI2_MISO
SPI2_MOSI
LEDPOL0
LEDPOL1
RUNLED
ERRLED
SYNC1
SYNC0
TXA_N
TXA_P
RXA_N
RXA_P
TXB_N
TXB_P
RXB_N
RXB_P
PORT_A
PORT_B
LED5
Emerald
R164 330R
R165 10K
LED6
RED
R166 330R
R167 10K
R168 10K
R169 10K
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Template\espark-consulting-group-800x300.jpg.
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Date: 8/8/2024
Page Num: 7
Drawn By: Huzaifa Ali
Page Total: 32

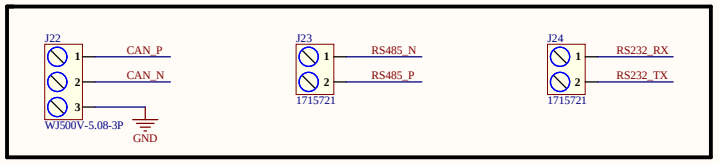
+24V OUTPUT



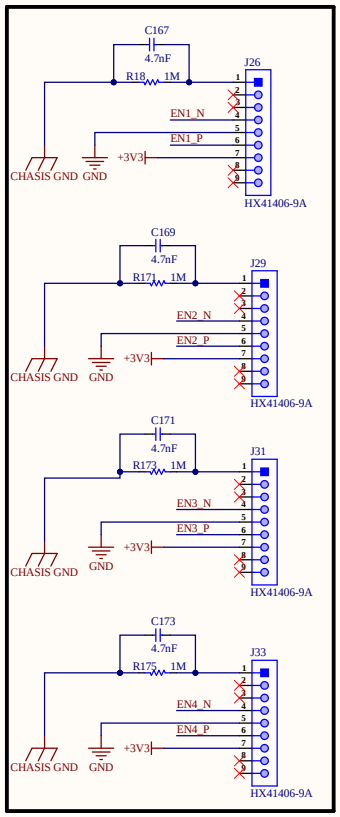
CAN

RS-485

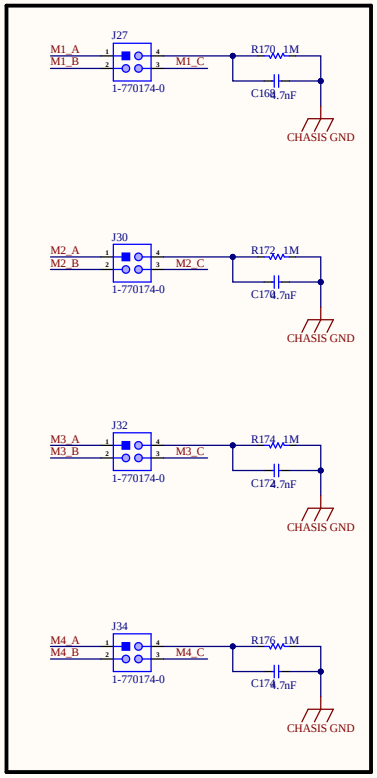
RS-232

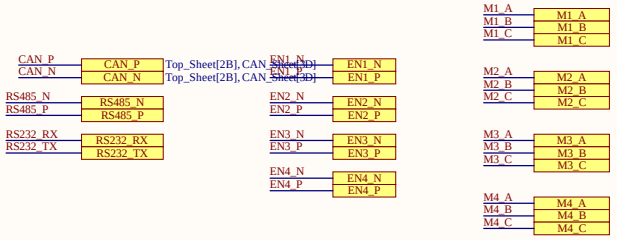


ENCODER



POWER PORT

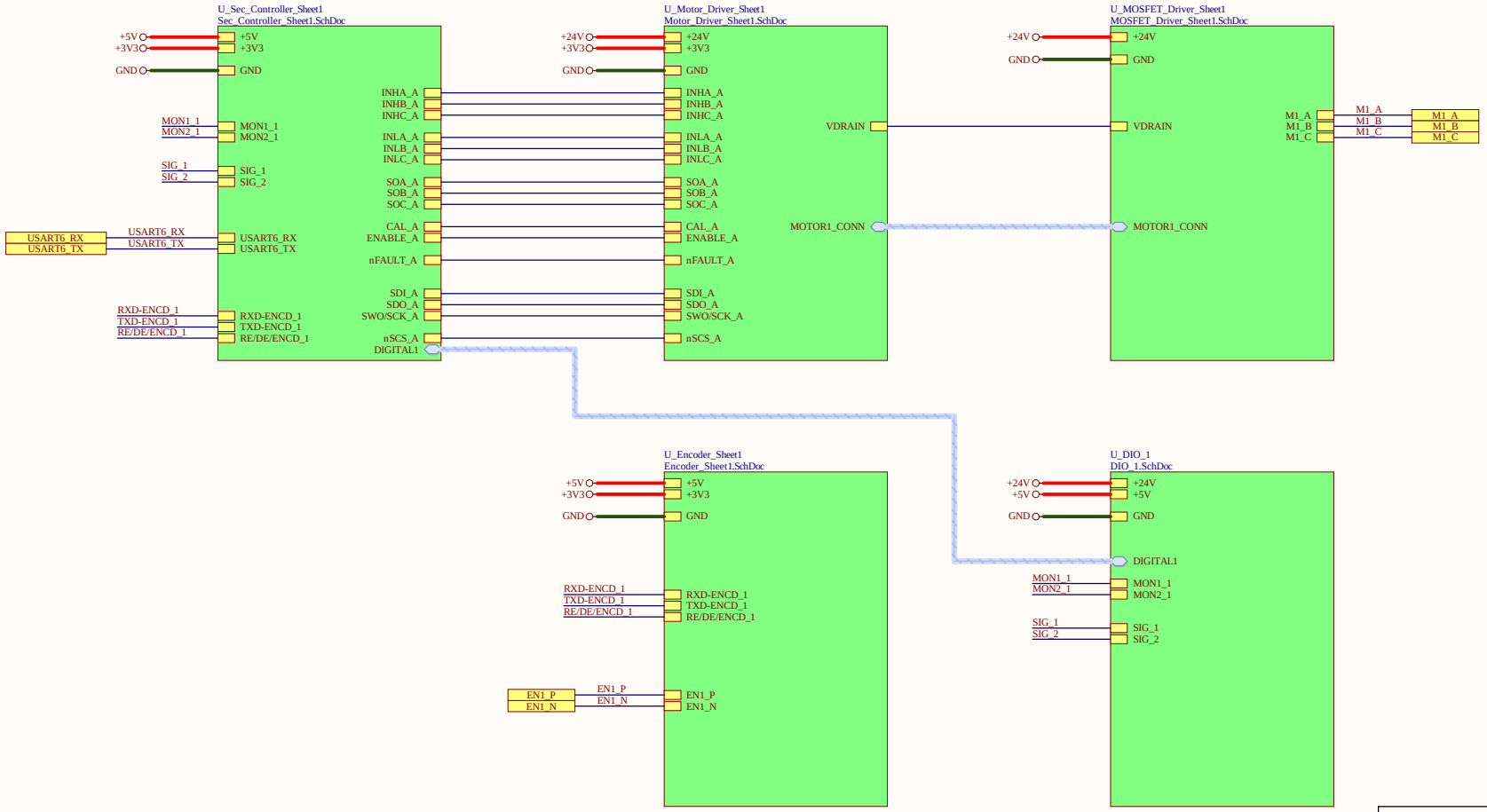






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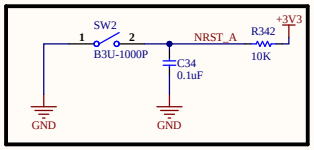
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Date: 8/9/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 32



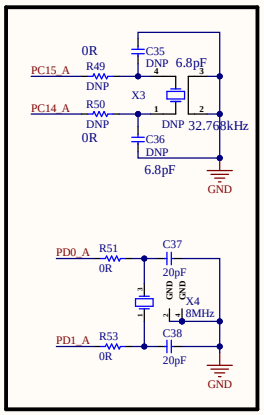
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File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 32

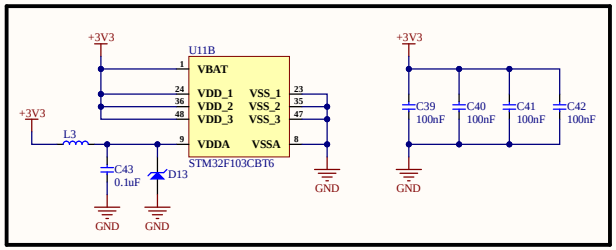
RESET



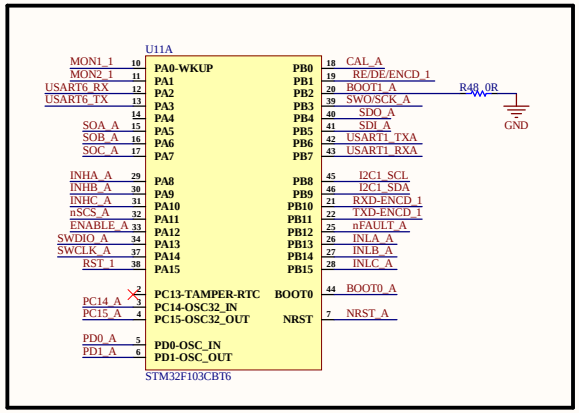
CRYSTALS



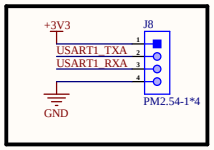
POWER



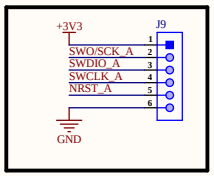
MCU



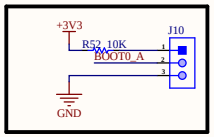
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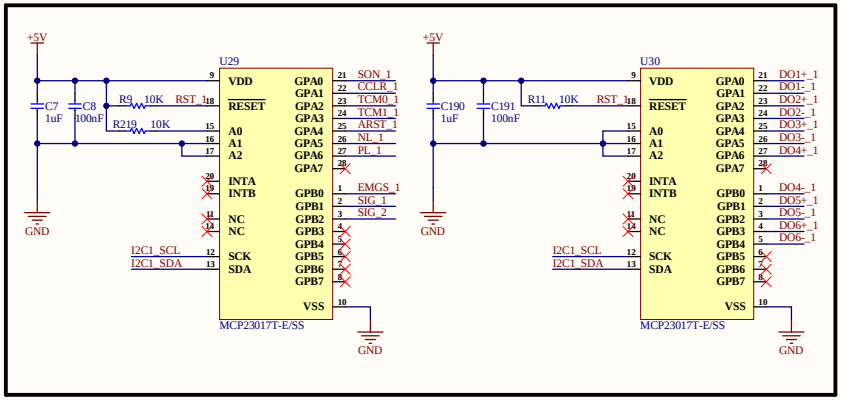
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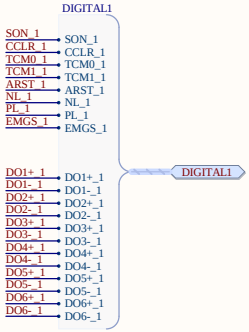


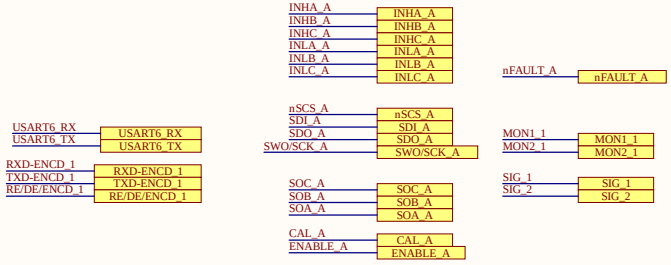
BOOT

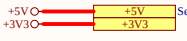


IO-EXPANDER





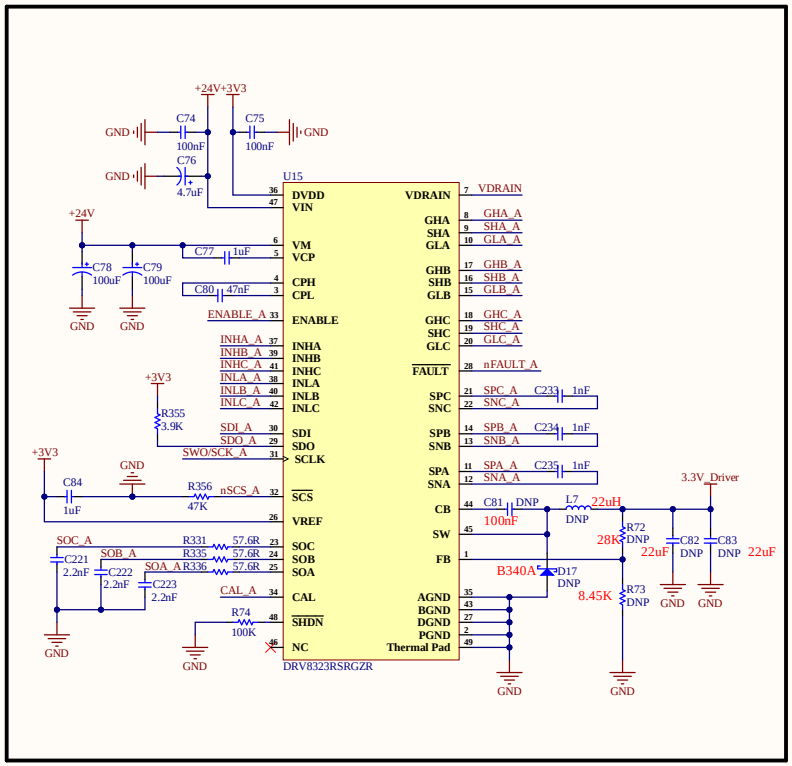


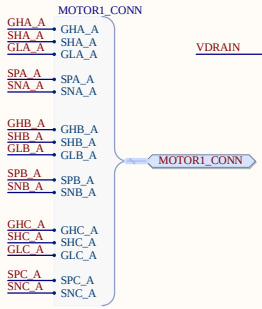


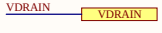
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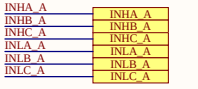
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Date: 8/8/2024	Page	Num: 10
Drawn By: Huzaifa Ali	Page	Total: 32

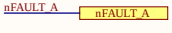
SERVO DRIVER

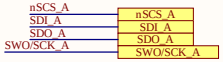


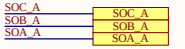


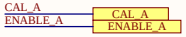


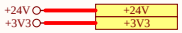








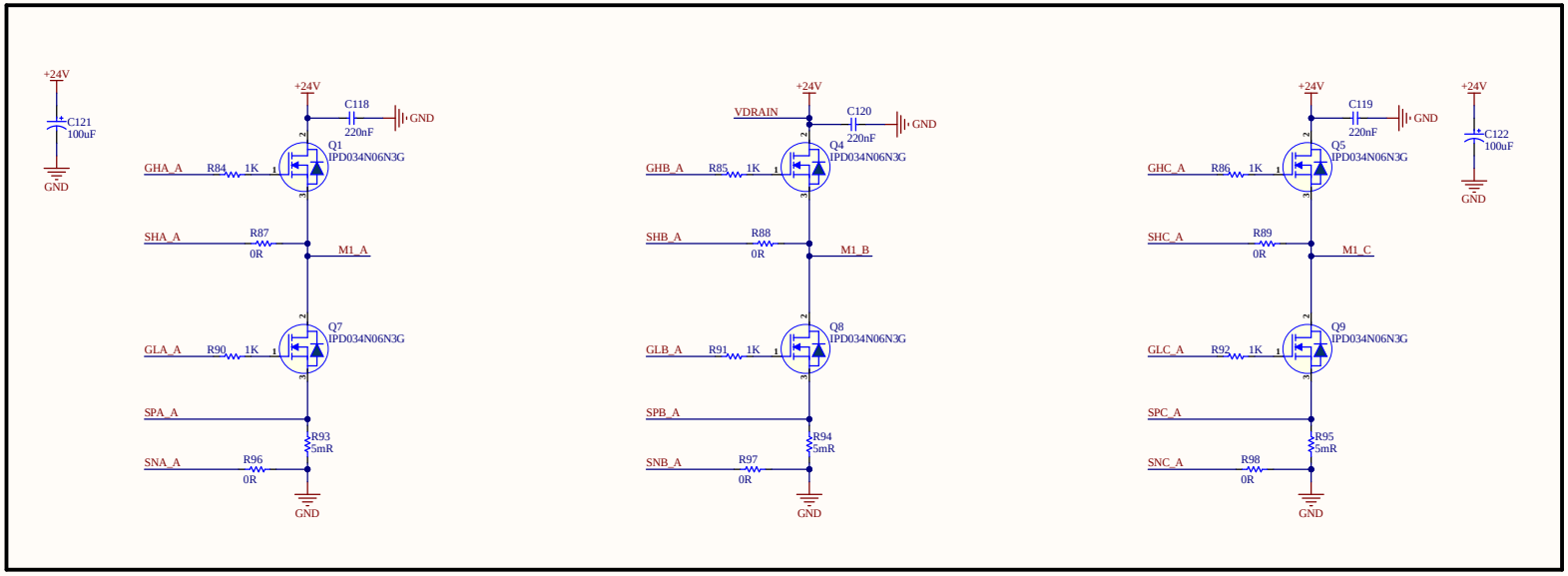




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File: Servo-Board-v1.0.PrjPcb		
Date: 8/8/2024	Page	Num: 11
Drawn By: Huzaifa Ali	Page	Total: 32

HALF BRIDGES



MOTOR1_CONN

GHA_A
SHA_A
GLA_A
SPA_A
SNA_A
GHB_A
SHB_A
GLB_A
SPB_A
SNB_A
GHC_A
SHC_A
GLC_A
SPC_A
SNC_A

MOTOR1_CONN

VDRAIN — VDRAIN

M1_A
M1_B
M1_C

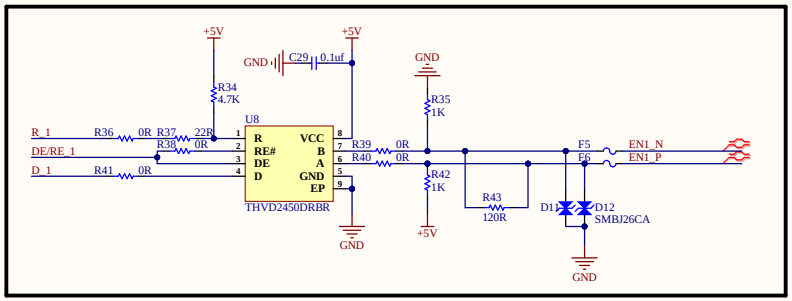
+24V — +24V

GND — GND

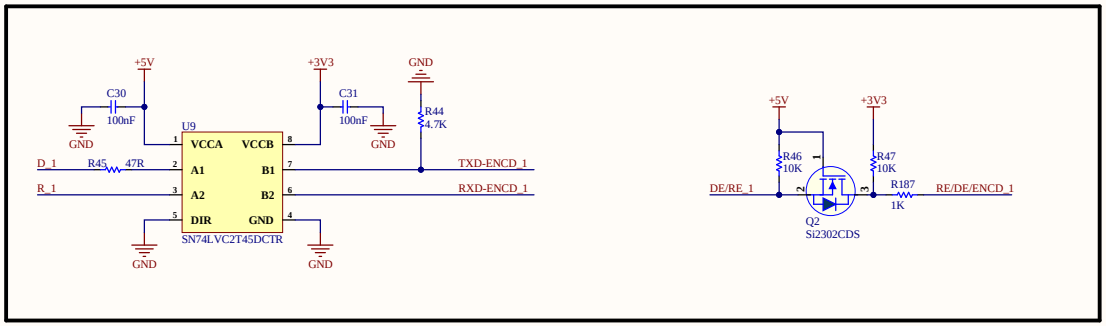
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File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 12
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ENCODER INPUT



LEVEL SHIFTER



EN1_N
EN1_P
EN1_N
EN1_P

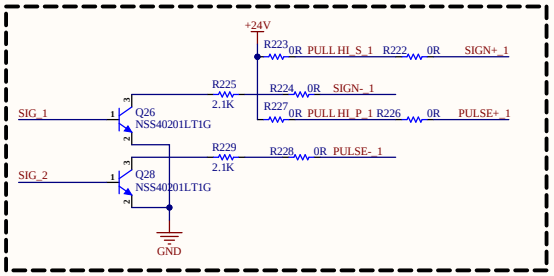
RXD-ENCD_1
TXD-ENCD_1
RE/DE/ENCD_1
RXD-ENCD_1
TXD-ENCD_1
RE/DE/ENCD_1

+5V
+3V3
GND
+5V
+3V3
GND

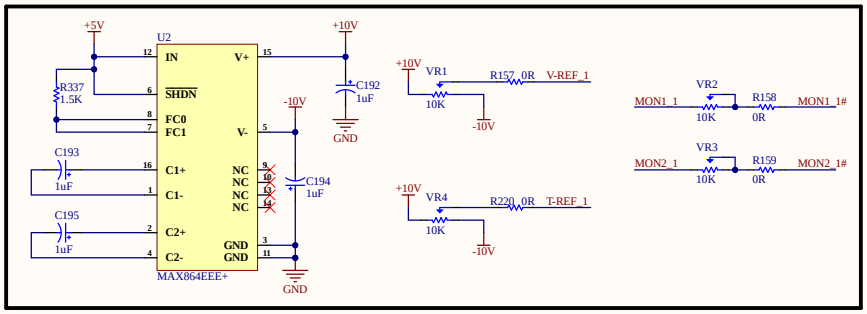
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Drawn By: Huzaifa Ali	Page Total: 32

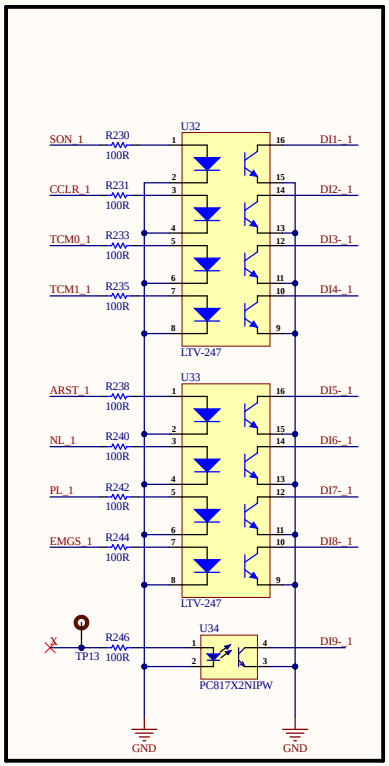
NPN



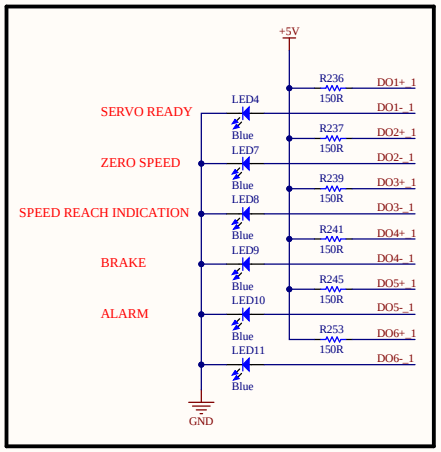
CHARGE PUMP



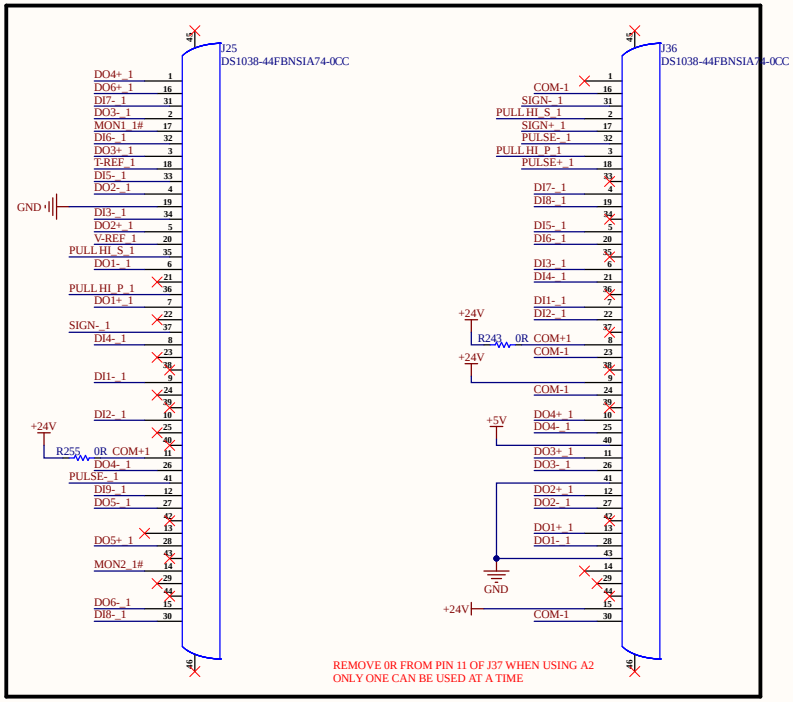
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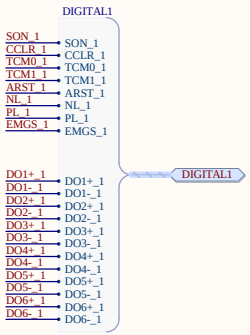


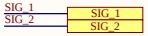
INDICATION LEDS

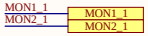


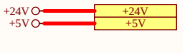
DB-44





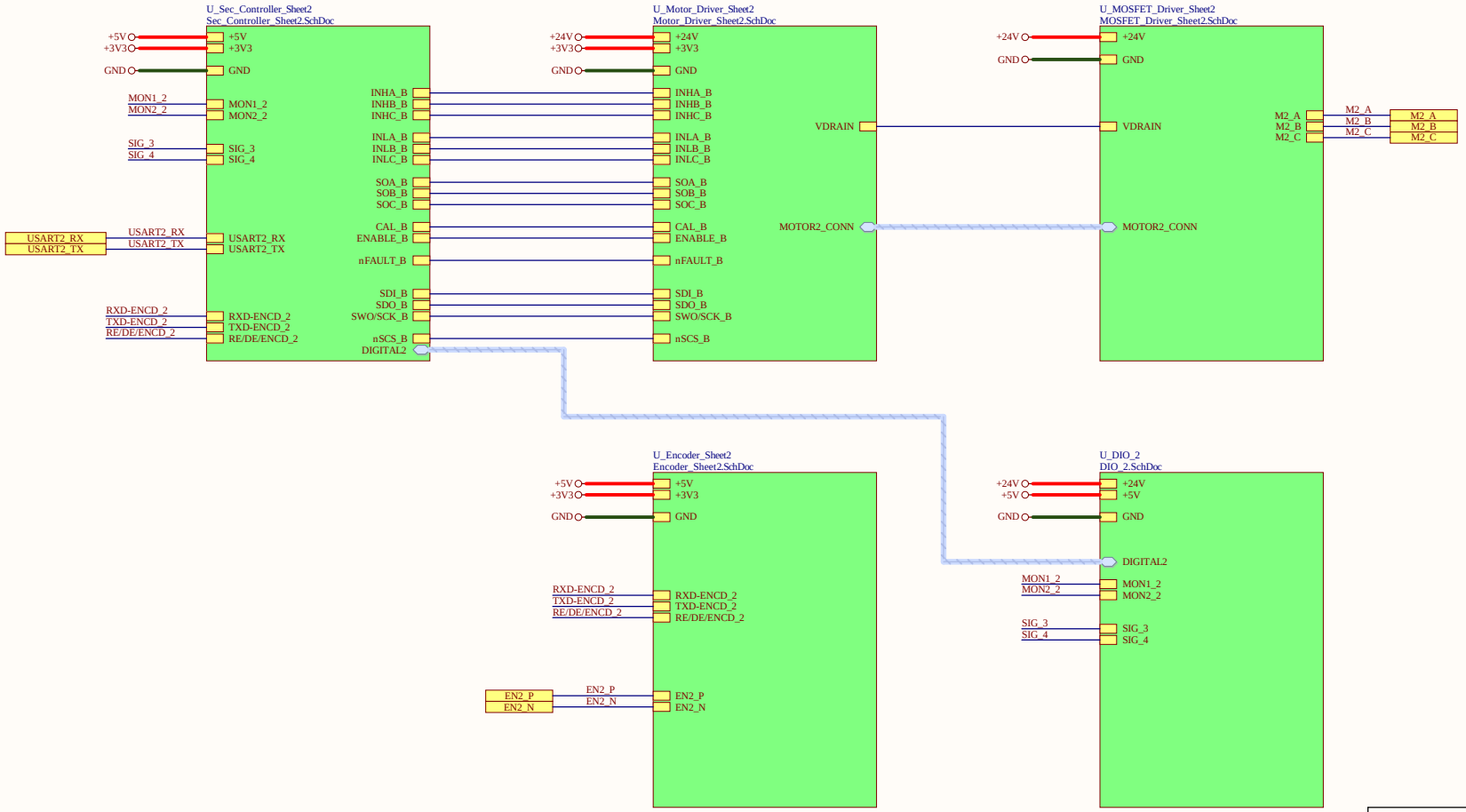






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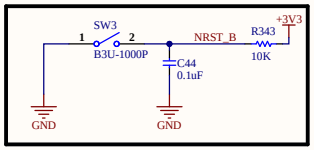
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Date: 8/8/2024	Page	Num: 14
Drawn By: Huzaifa Ali	Page	Total: 32



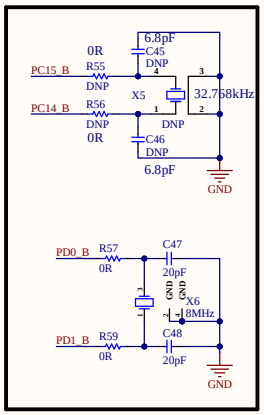
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File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 15
Drawn By: Huzaifa Ali	Page Total: 32

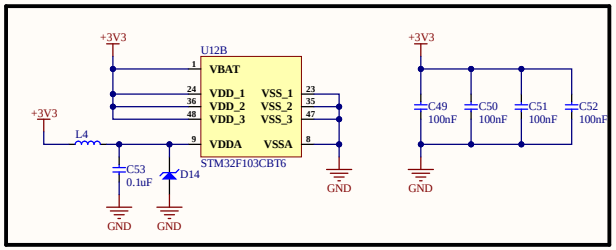
RESET



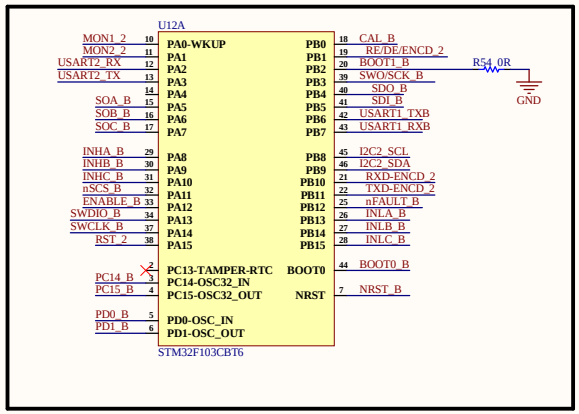
CRYSTALS



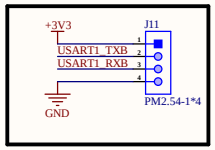
POWER



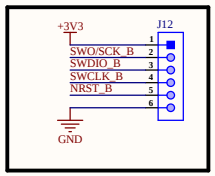
MCU



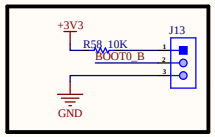
UART



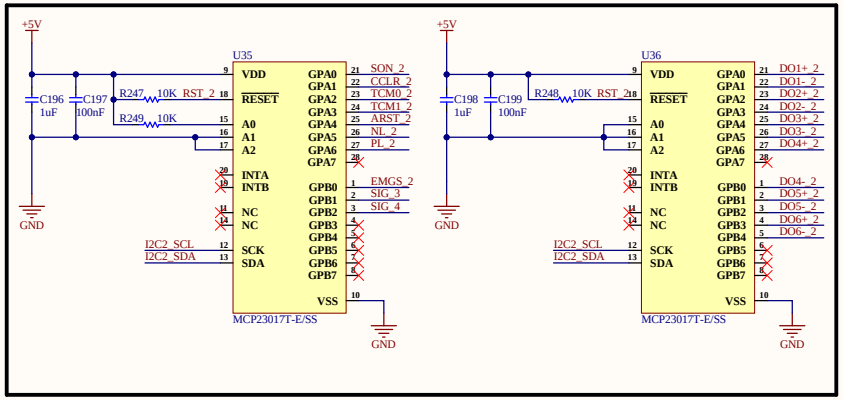
SWD PORT

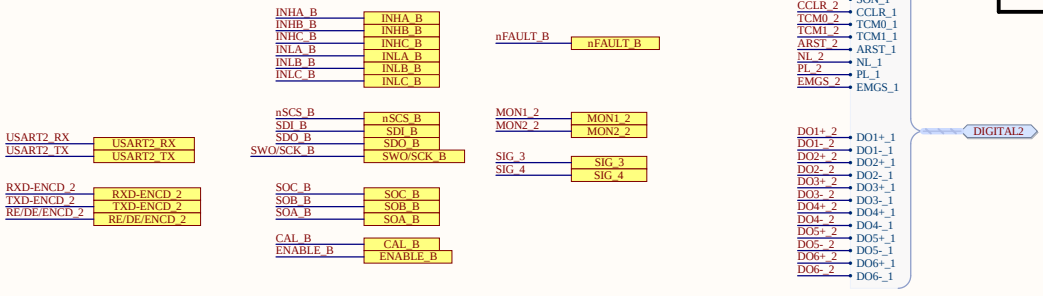


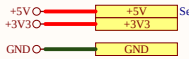
BOOT



IO-EXPANDER



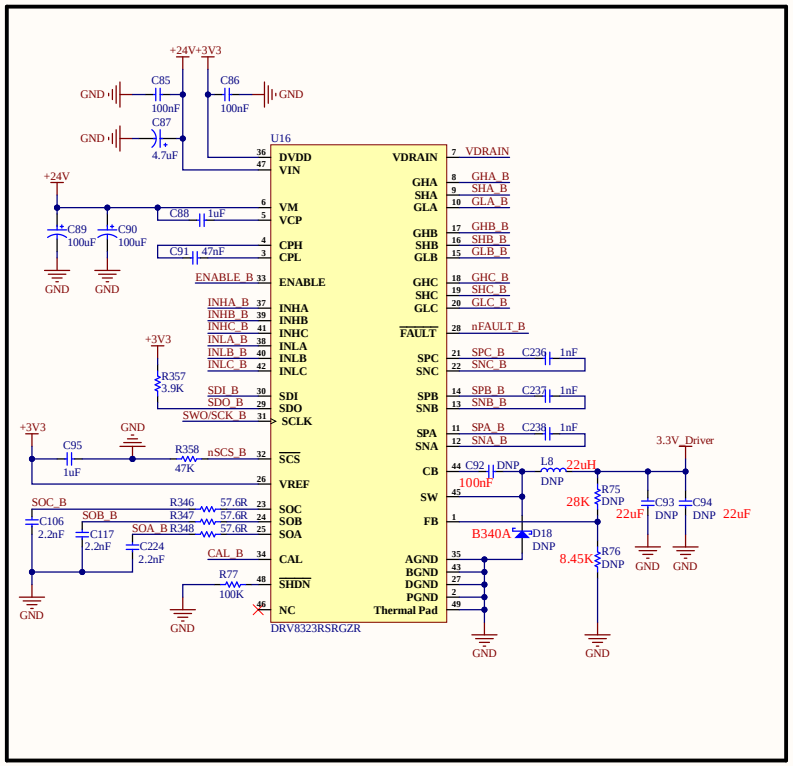


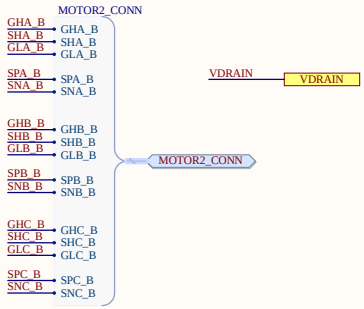


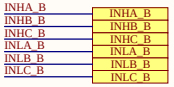
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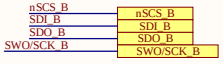
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Date: 8/8/2024	Page	Num: 16
Drawn By: Huzaifa Ali	Page	Total: 32

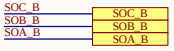
SERVO DRIVER

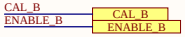


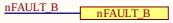


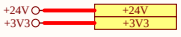








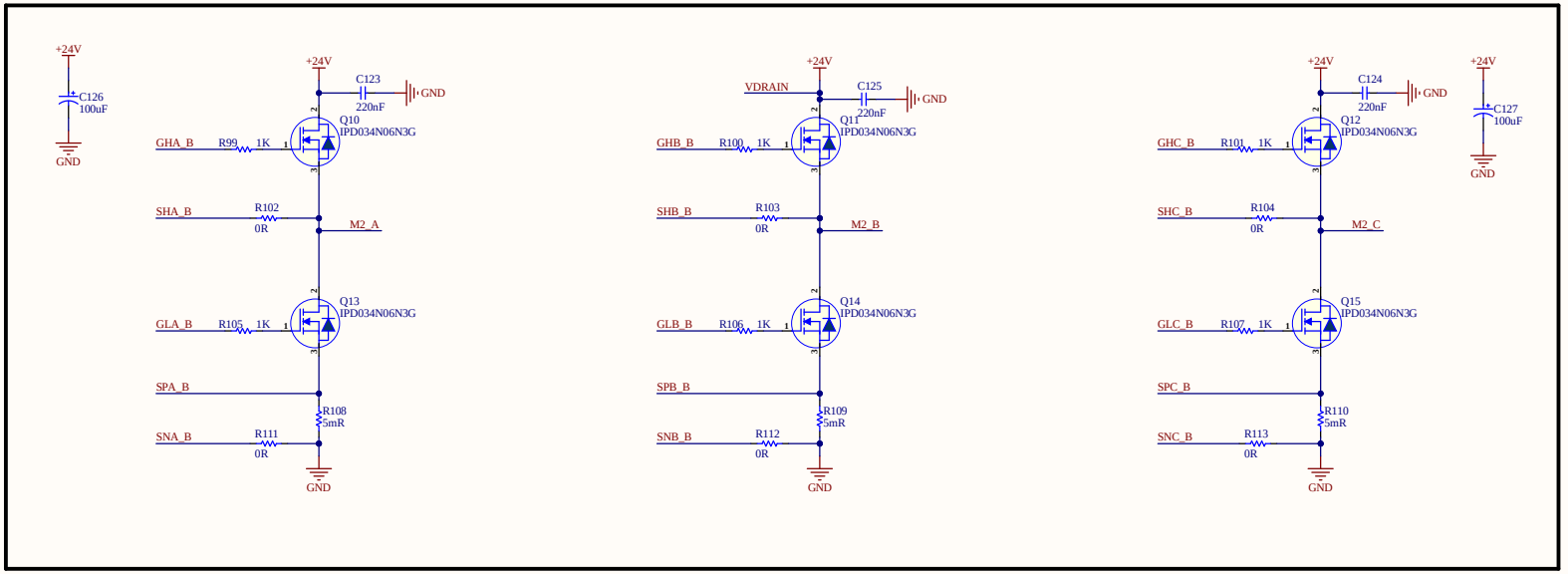




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Date: 8/8/2024	Page	Num: 17
Drawn By: Huzaifa Ali	Page	Total: 32

HALF BRIDGES



MOTOR2_CONN

GHA_B — GHA_B
SHA_B — SHA_B
GLA_B — GLA_B
SPA_B — SPA_B
SNA_B — SNA_B
GHB_B — GHB_B
SHB_B — SHB_B
GLB_B — GLB_B
SPB_B — SPB_B
SNB_B — SNB_B
GHC_B — GHC_B
SHC_B — SHC_B
GLC_B — GLC_B
SPC_B — SPC_B
SNC_B — SNC_B

MOTOR2_CONN

VDRAIN — VDRAIN

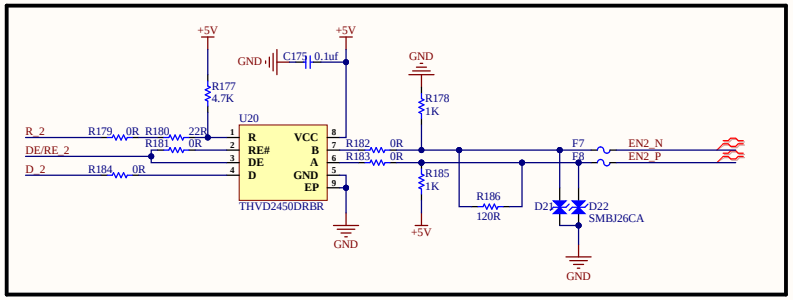
M2_A — M2_A
M2_B — M2_B
M2_C — M2_C

+24V — +24V
GND — GND

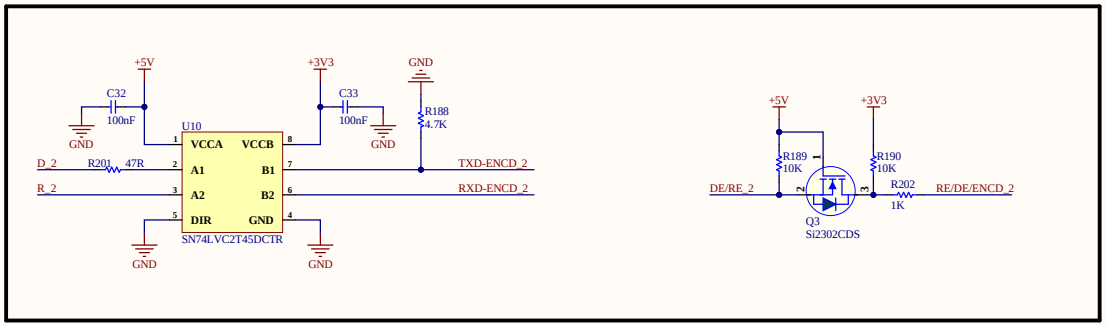
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D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: MOSFET_Driver_Sheet2.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 18
Drawn By: Huzaifa Ali	Page Total: 32

ENCODER INPUT



LEVEL SHIFTER



EN2_N
EN2_P
EN2_N
EN2_P

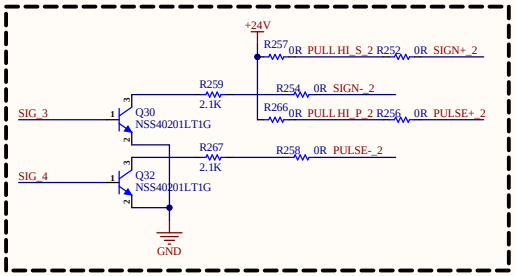
RXD-ENCD_2
TXD-ENCD_2
RE/DE/ENCD_2
RXD-ENCD_2
TXD-ENCD_2
RE/DE/ENCD_2

+5V
+3V3
GND
+5V
+3V3
GND

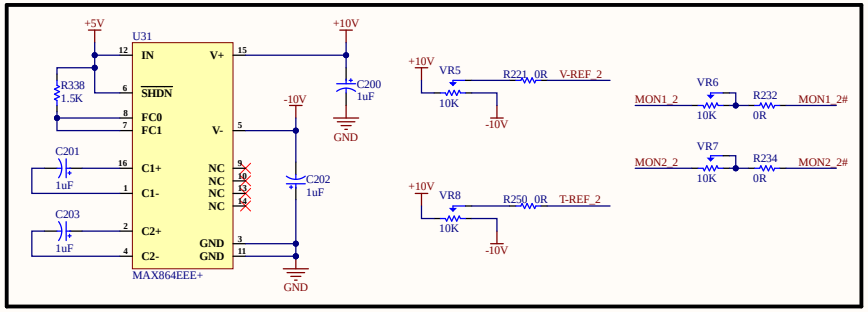
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Title: Encoder_Sheet2.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 19
Drawn By: Huzaifa Ali	Page Total: 32

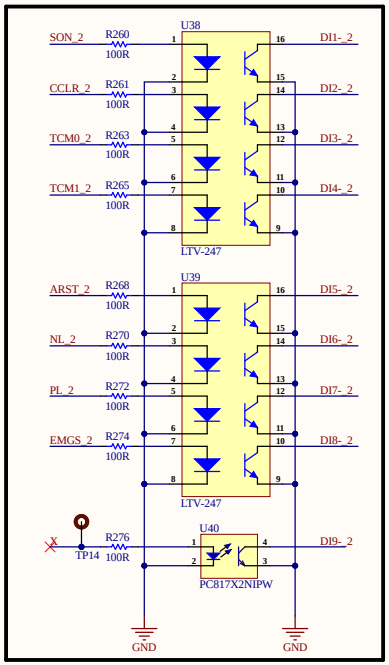
NPN



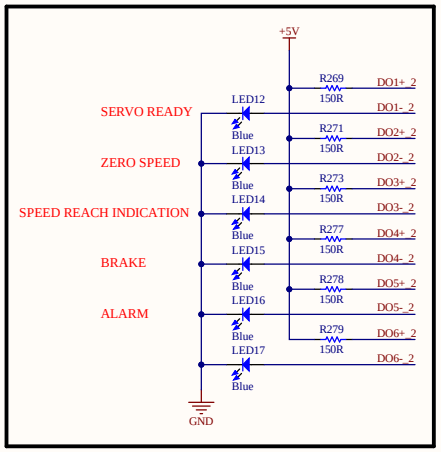
CHARGE PUMP



DIO INPUTS

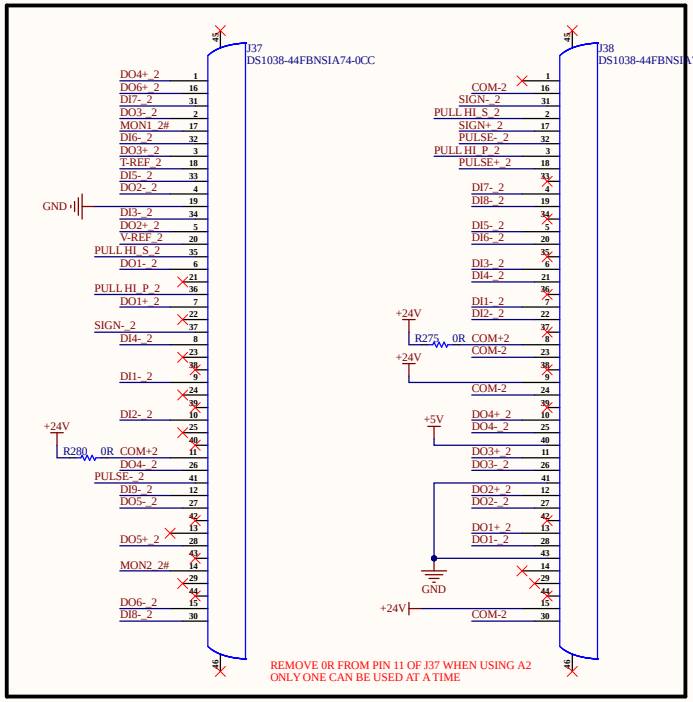


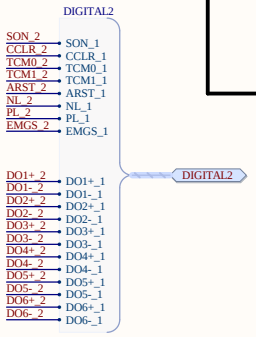
INDICATION LEDS

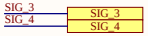


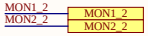
CATEGORY B

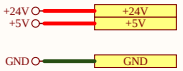
CATEGORY A2





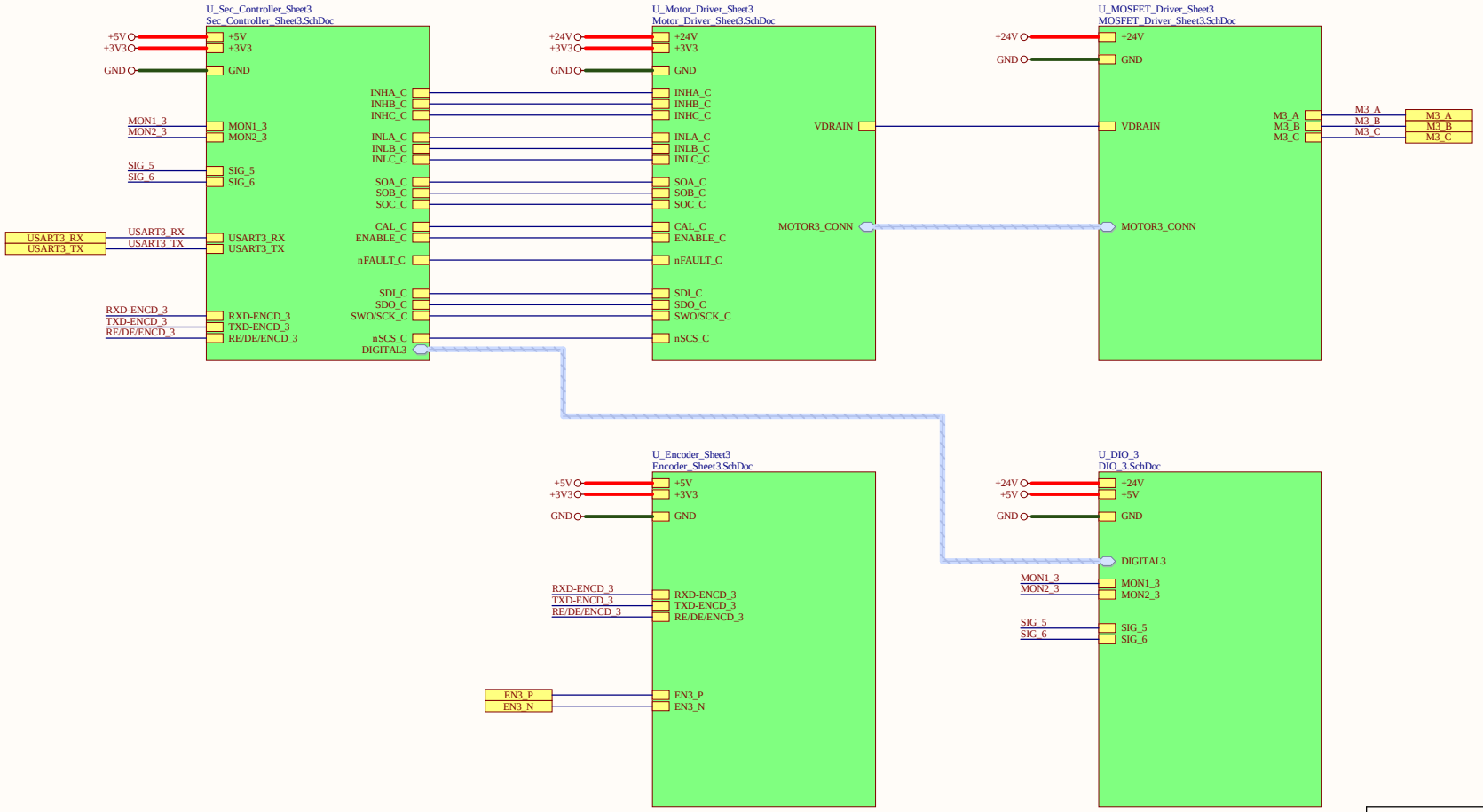






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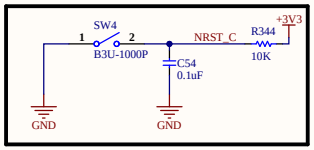
Title: DIO_2.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 20
Drawn By: Huzaifa Ali	Page Total: 32



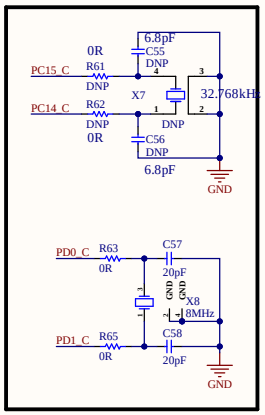
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File does not exist.

Title: Servo3.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 21
Drawn By: Huzaifa Ali	Page Total: 32

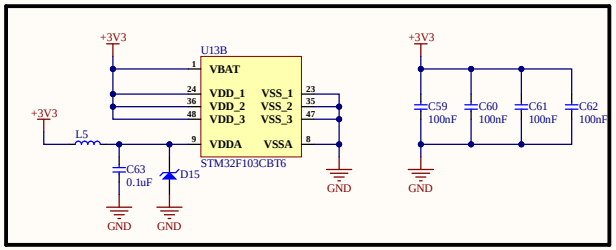
RESET



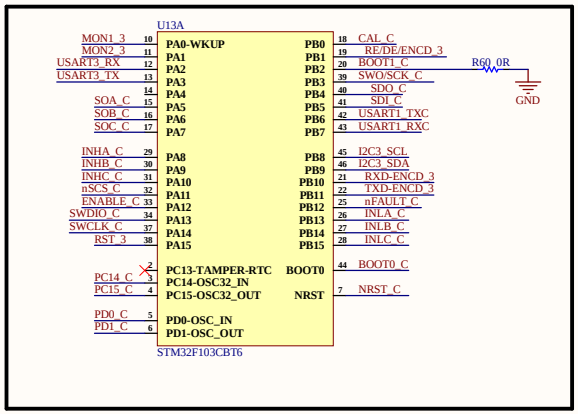
CRYSTALS



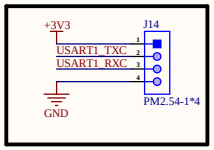
POWER



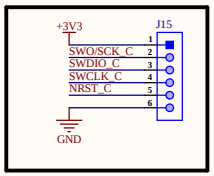
MCU



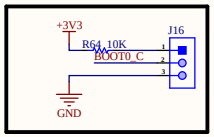
UART



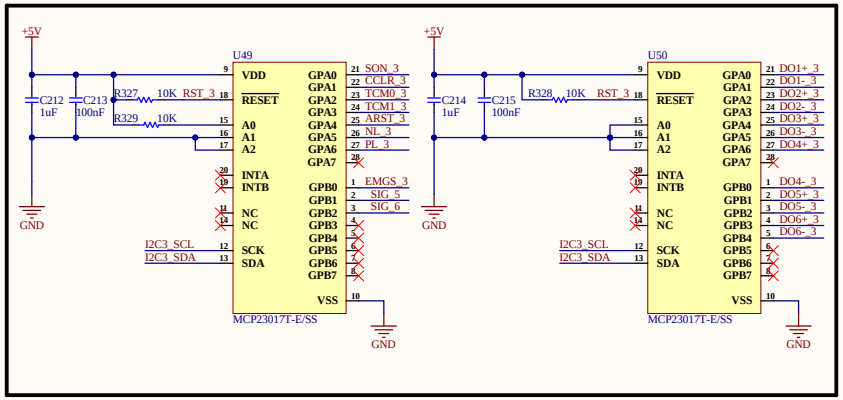
SWD PORT



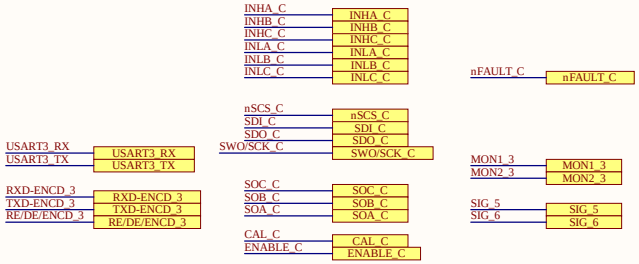
BOOT

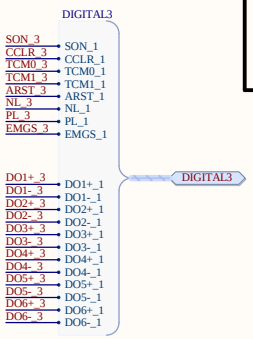


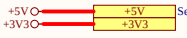
IO-EXPANDER



Hierarchical Sheet Entries



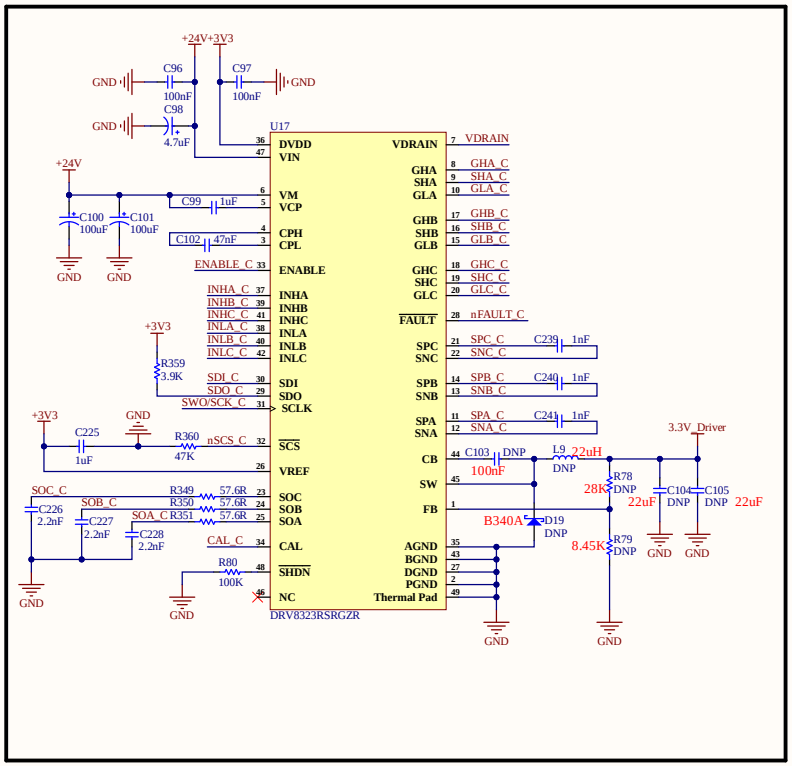


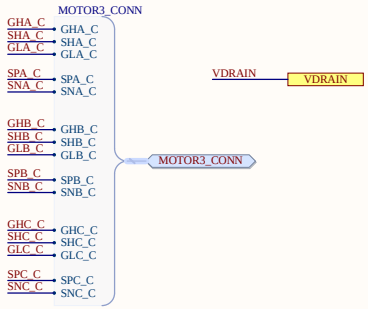


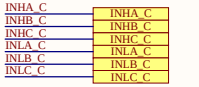
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File does not exist.

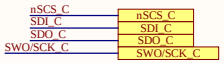
Title: Sec_Controller_Sheet3.SchDoc		
File: Servo-Board-v1.0.PrjPcb		
Date: 8/8/2024	Page	Num: 22
Drawn By: Huzaifa Ali	Page	Total: 32

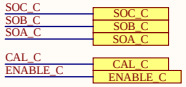
SERVO DRIVER

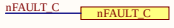










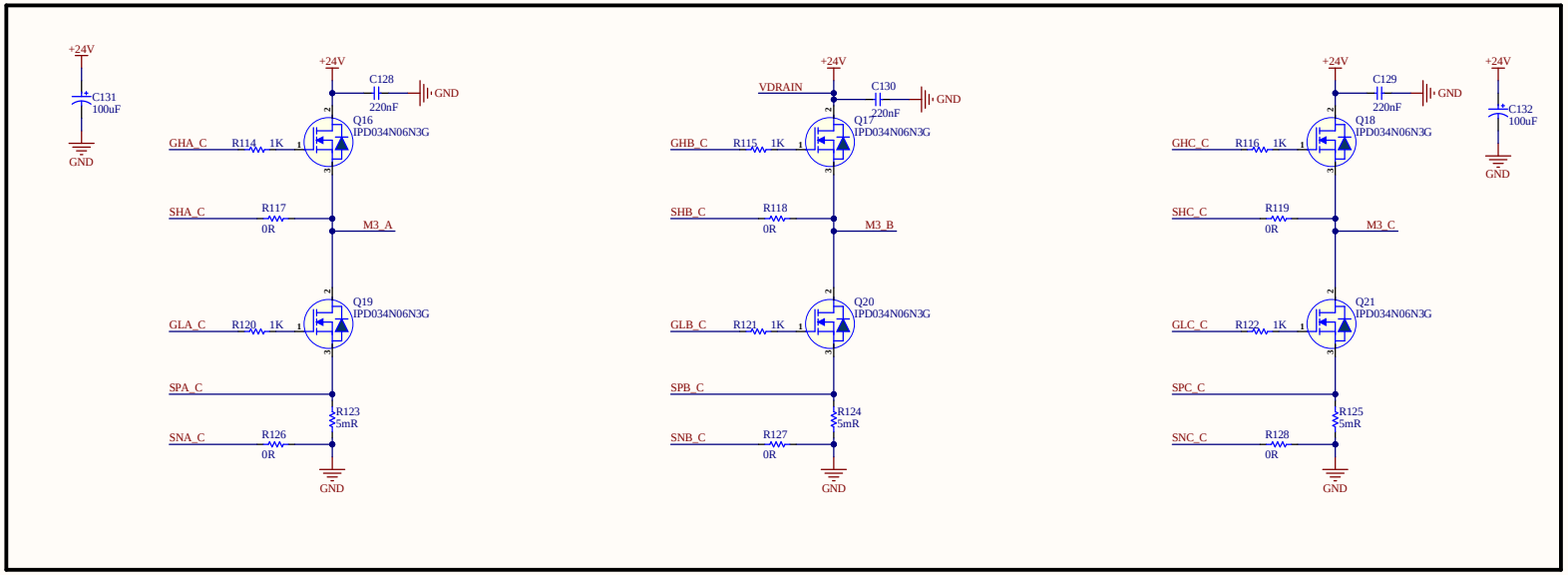


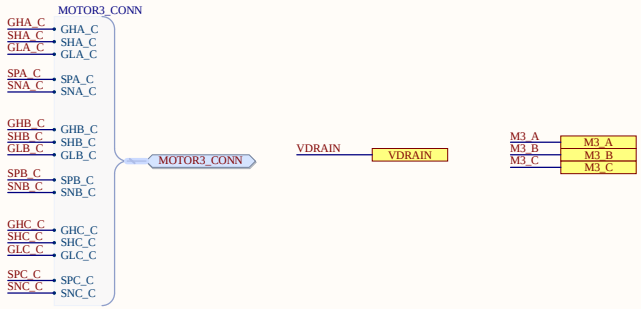


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Title: Motor_Driver_Sheet3.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 23
Drawn By: Huzaifa Ali	Page Total: 32

HALF BRIDGES



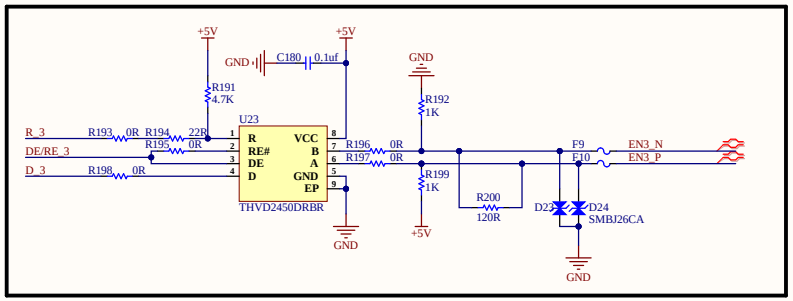




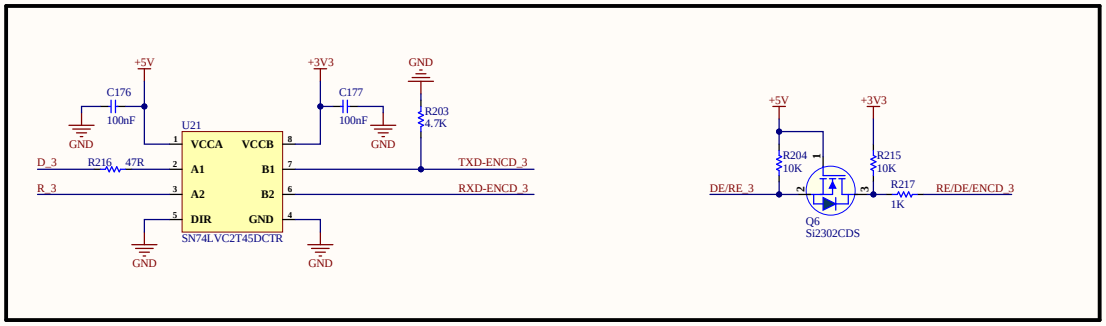
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Title: MOSFET_Driver_Sheet3.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 24
Drawn By: Huzaifa Ali	Page Total: 32

ENCODER INPUT



LEVEL SHIFTER



EN3_N
EN3_P
EN3_N
EN3_P

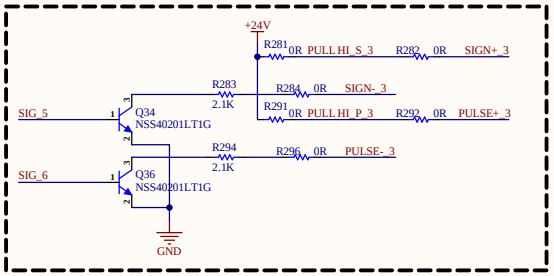
RXD-ENCD_3
TXD-ENCD_3
RE/DE/ENCD_3
RXD-ENCD_3
TXD-ENCD_3
RE/DE/ENCD_3

+5V
+3V3
GND
+5V
+3V3
GND

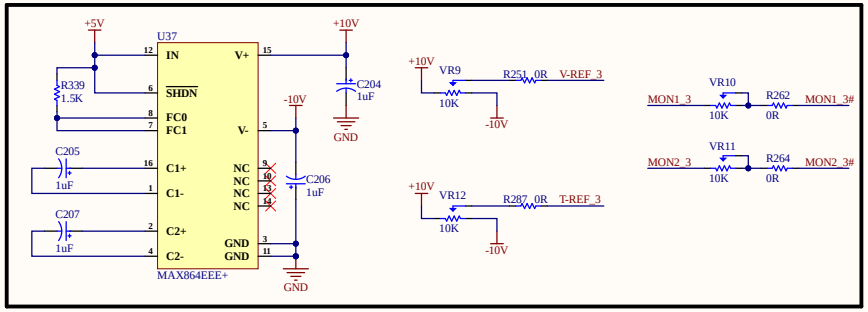
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Encoder_Sheet3.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 25
Drawn By: Huzaifa Ali	Page Total: 32

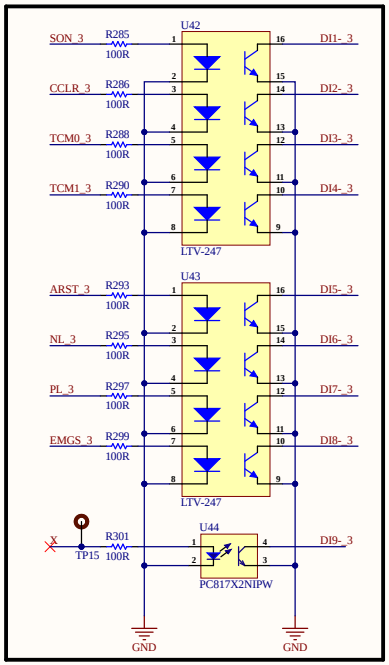
NPN



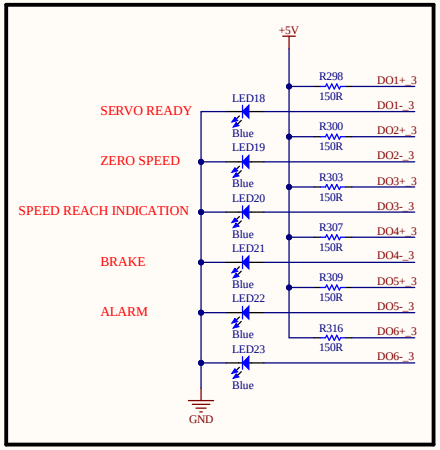
CHARGE PUMP



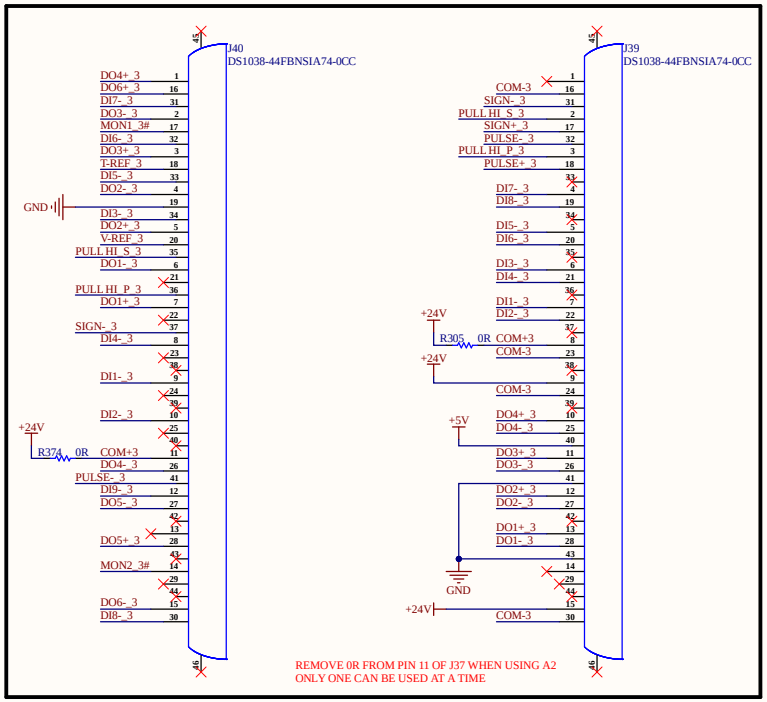
DIO INPUTS

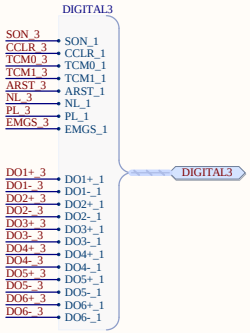


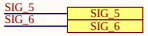
INDICATION LEDS

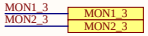


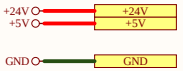
DB-44





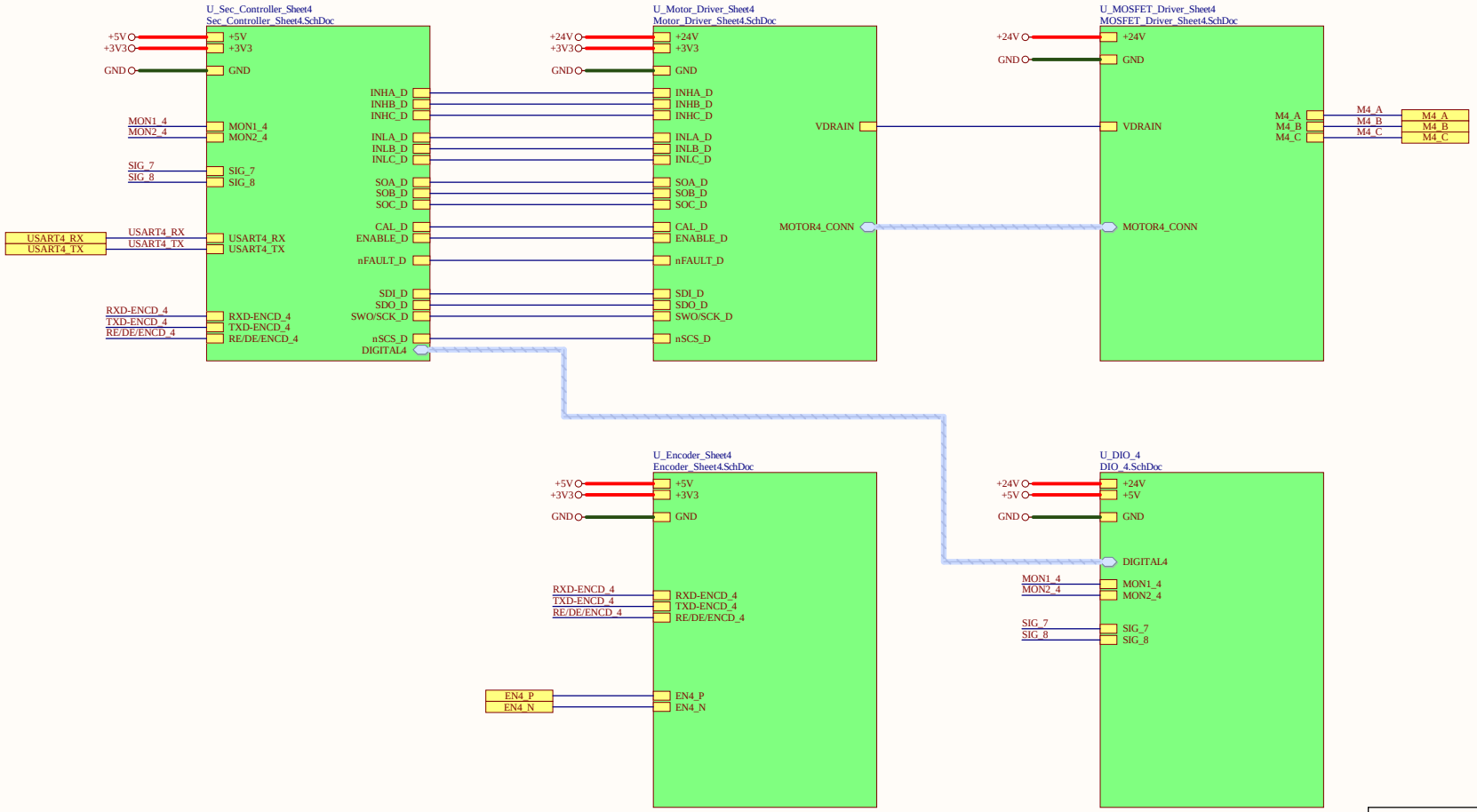






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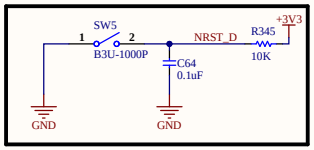
Title: DIO_3.SchDoc		
File: Servo-Board-v1.0.PrjPcb		
Date: 8/8/2024	Page Num:	26
Drawn By: Huzaifa Ali	Page Total:	32



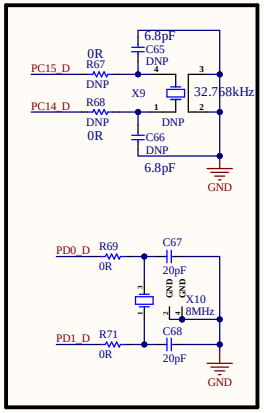
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Servo4.SchDoc	
File: Servo-Board-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 27
Drawn By: Huzaifa Ali	Page Total: 32

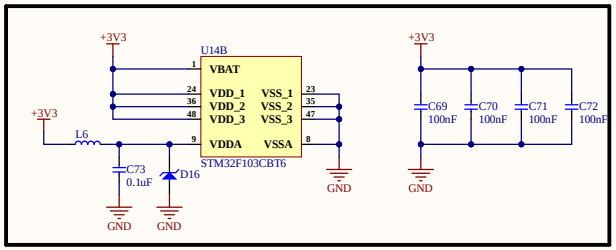
RESET



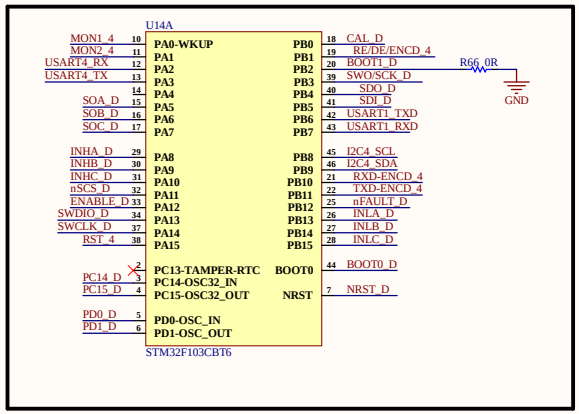
CRYSTALS



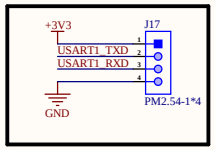
POWER



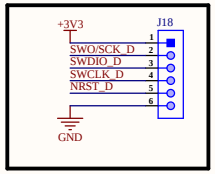
MCU



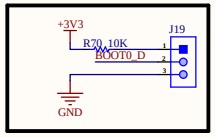
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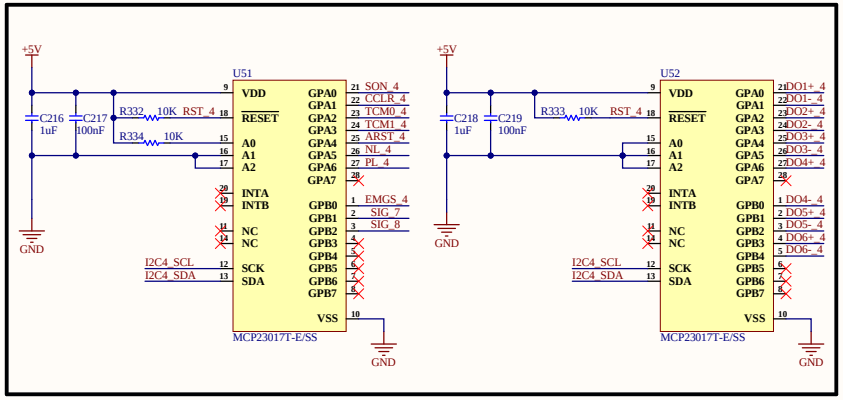
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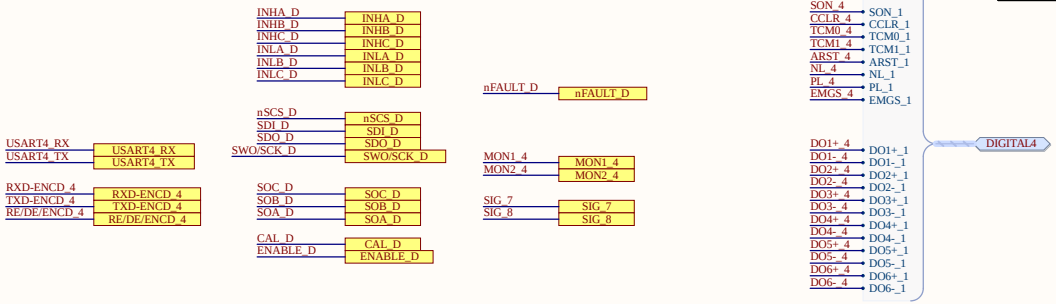


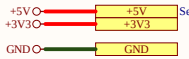
BOOT



IO-EXPANDER



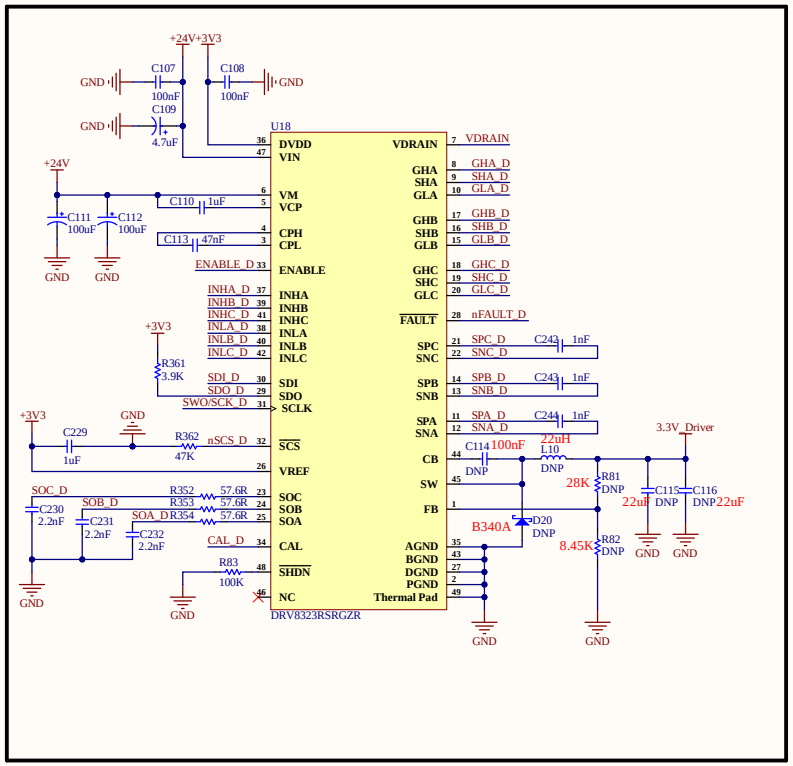


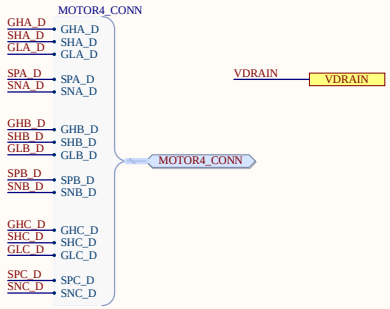


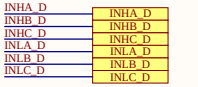
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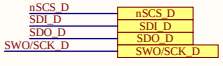
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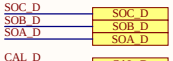
SERVO DRIVER

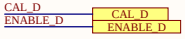


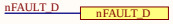


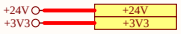










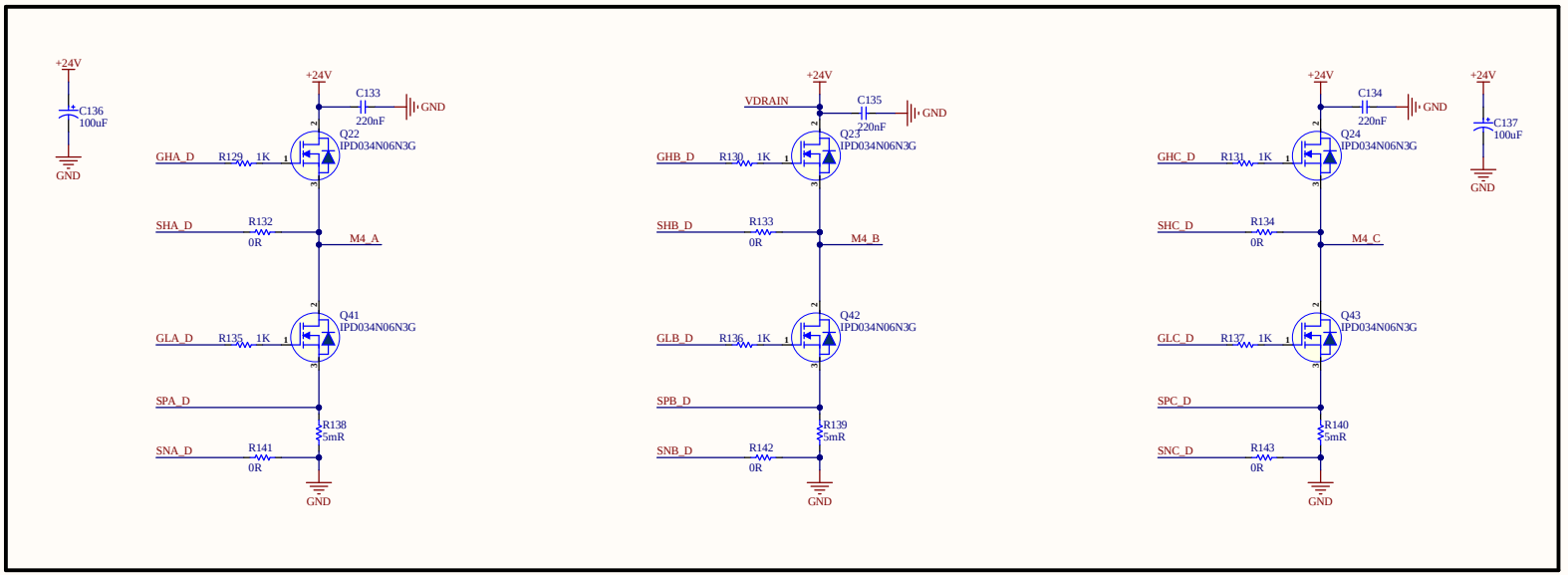


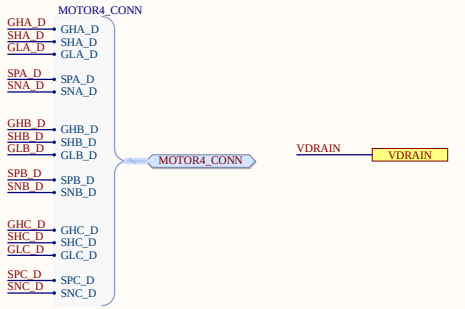


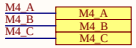
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HALF BRIDGES





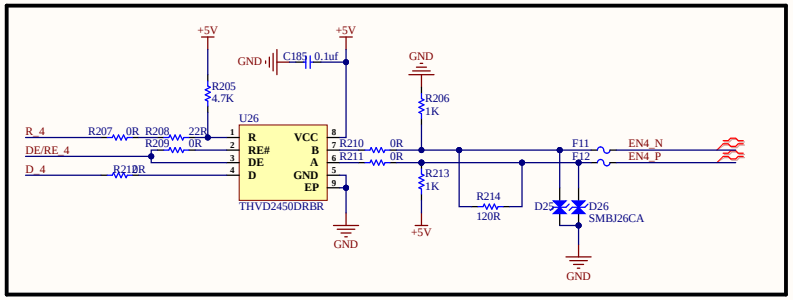




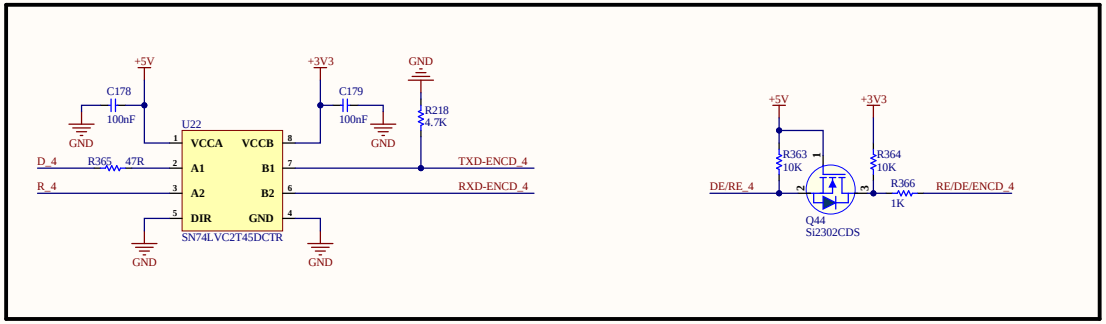
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ENCODER INPUT



LEVEL SHIFTER



EN4_N
EN4_P
EN4_N
EN4_P

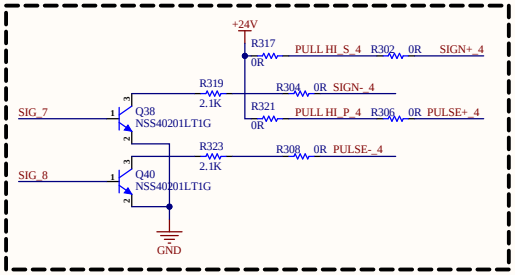
RXD-ENCD_4
TXD-ENCD_4
RE/DE/ENCD_4
RXD-ENCD_4
TXD-ENCD_4
RE/DE/ENCD_4

+5V
+3V3
GND
+5V
+3V3
GND

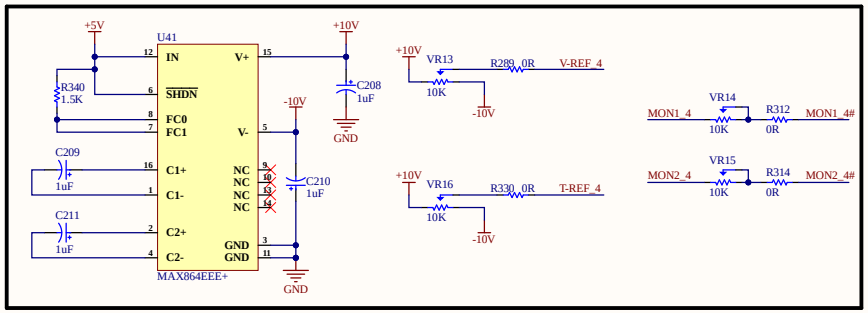
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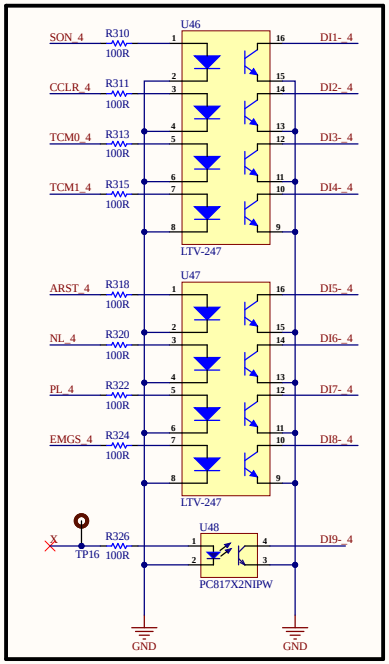
NPN



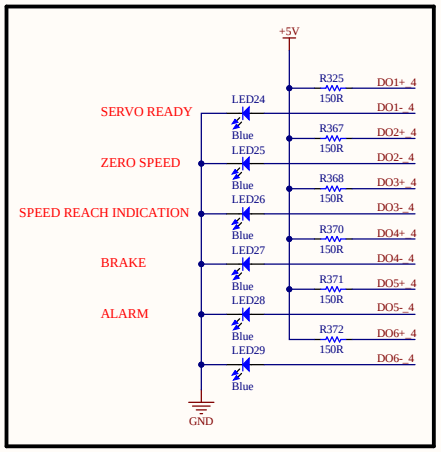
CHARGE PUMP



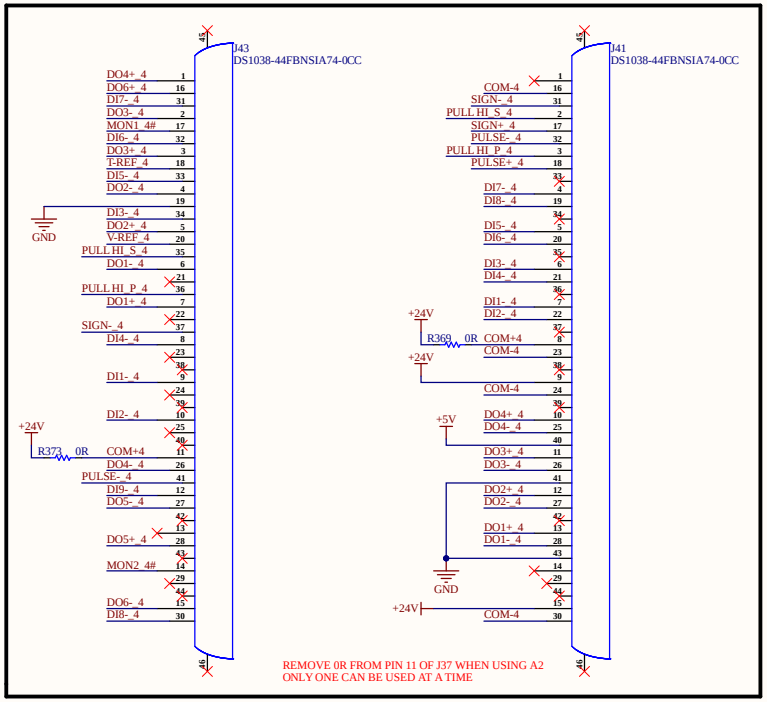
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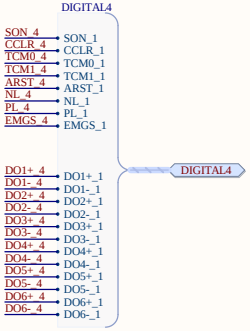


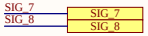
INDICATION LEDS

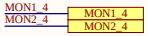


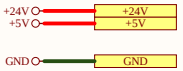
DB-44











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